

LUMABANK ACTIVATION EXPERIMENT

Project Documentation

ETL · data model · DAX · report · build walkthrough

Fictional company · synthetic seeded data · generated 2026-09-11

00 — Scope & Scaffold (Phase 0)

Project: LumaBank — New-User Activation & the Onboarding Experiment That Broke Mid-Flight **Fictional company:** LumaBank — a Brazilian neobank (digital account + card). **Randomization:** per **user**, 50/50, via a feature-flag service. **Primary metric:** **Day-7 Activation** — account created **and** first transaction within 7 days. **Date locked:** 2026-09-10

PT — resumo. Projeto de portfólio nº 5, e o primeiro de **fintech**. A LumaBank é um neobank brasileiro fictício (conta digital + cartão), moeda **BRL apenas**. O time de Growth/Produto redesenhou o fluxo de **onboarding/KYC** (menos etapas, upload de documento adiado para *depois* da criação da conta) para aumentar a ativação de novos usuários. A métrica primária é **Day-7 Activation** (conta criada e primeira transação até o 7º dia). O experimento é **randomizado por usuário**, 50/50, com janela planejada de **6 semanas**. Por volta do **dia 11**, o monitoramento diário de **SRM (sample ratio mismatch)** detecta um desvio significativo da proporção 50/50 — que começa **exatamente no dia de um deploy do app mobile** sem relação com o experimento. A investigação precisa separar dois efeitos do bug (contaminação de quem já estava no teste × viés de atribuição dos novos cadastros, correlacionado com região/timezone), avaliar três alternativas de correção com trade-offs de viés × poder × prazo, e refazer o experimento numa **janela mais curta** com o **poder/MDE recalculado**. Não é a história de "rodei um A/B test e deu certo" — é a de um experimento que quebra em produção e é diagnosticado e refeito sob prazo.

1. Premise

LumaBank is a Brazilian neobank: users open a free digital checking account and a card entirely in-app. Growth is healthy at the top of the funnel — paid social, referral and app-store search all bring signups — but a large share of new users **never transact**. They download the app, start the sign-up, hit the KYC step (a full identity-document upload *before* the account exists), and drop. The users who do get through often take more than a week to make a first payment, by which point intent has cooled.

The Growth and Product teams redesigned the onboarding flow: **fewer steps to a working account, and the identity-document upload deferred to *after* account creation** (the account opens in a limited state, and the document is requested before limits are lifted or a physical card ships). The hypothesis is that getting a usable account into the user's hands sooner converts early intent into a first transaction.

This was launched as a **randomized experiment, unit = user, 50/50**, through the feature-flag service: on assignment a new user sees either the **new** onboarding (treatment) or the **existing** one (control). Planned window: **6 weeks**. Primary metric: **Day-7 Activation**.

What makes this project a portfolio piece is what happened next. Around **day 11**, the daily **sample-ratio-mismatch (SRM)** check — a standing part of how LumaBank runs any experiment — flagged that the arms were no longer 50/50, and the drift had started on the exact day of an unrelated **mobile app deploy**. The rest of the project is the incident: detection → root-cause diagnosis → a documented decision between three imperfect fixes under business time pressure → statistical re-planning → a clean re-run.

2. Locked scope decisions

Dimension	Decision
Deliverable language	English (all docs, code, reports). Every deliverable also carries a short PT explanation of each step. Working chat in Portuguese.
Priority	Phase 10 (data_analysis/) is the primary deliverable — the experiment incident, end to end. Power BI is secondary, scoped to 4 pages . Follows the AeroVanti model (analysis-primary, PBI lite), not VoltEdge/Voxa+/SwiftBite.
Portfolio gap this fills	No other project covers fintech , and none covers an experiment that breaks mid-flight on a real production bug and is diagnosed and re-run under deadline.
Geography / currency	Brazil only, BRL only . No second country, no FX — that ground is covered by VoltEdge.
Data window	21 months: 2025-01-01 → 2026-09-30 . Months 1–17 give the pre-experiment activation trend; the original experiment sits in month 19; the re-run finishes inside month 21 (with room for its Day-7 outcomes).
Original experiment window	6 weeks planned: 2026-07-06 → 2026-08-16 . Terminated early.
The break	Mobile app deploy on 2026-07-16 (experiment day 11) resets the local variant-assignment cache for a subset of users. Hotfix release 2026-07-22 .
Re-run window	4 weeks: 2026-08-24 → 2026-09-20 , bug fixed, power/MDE recomputed for the smaller sample the shorter window allows. The business could not wait another full 6 weeks.
Platform scale	~130,000 signups over the 21 months; end run-rate ~2,200–2,600 signups/week ; ~60,000 active users (≥1 transaction in the trailing 30 days) at window end.
Experiment scale	Original: ~13,500 randomized users (~6,750/arm). Re-run: ~9,000 (~4,500/arm).
ETL / data-quality depth	Balanced (matches the other four): 6 simulated source systems, 7 generic data-quality issue classes (\$5) plus a bug-specific catalog (\$6).
Power BI track	Semantic model as code (TMDL) + PBIR report + <code>theme.json</code> + build script; 4 pages , one business question per page.
Visual identity	New palette, designed in Phase 5 — must not reuse AeroVanti (navy/amber), VoltEdge (navy), or Voxa+ (violet/amber). Direction: trust/fintech without the money-green cliché.

3. Phase-1 report pages (4 pages, one question each)

#	Page	The question it answers
1	Executive Overview	Is new-user activation healthy and trending the right way, and what did the (re-run) onboarding experiment conclude?
2	Onboarding & Activation Funnel	Where do new users drop — signup → KYC → first transaction — and how does the funnel differ between the old flow and the new one?
3	Experiment Integrity Monitor	Can the experiment be trusted? Daily allocation by arm, SRM p-value over time, the deploy/break marked, contaminated vs clean population. <i>This page exists to tell the break story visually.</i>
4	Experiment Readout	Did the new onboarding work on the re-run — primary effect with CI, guardrails (KYC rejection / fraud) held, and does the effect differ by acquisition channel?

4. Experiment design (locked at Phase 0, detailed in Phase 1)

4.1 The original experiment (as pre-registered)

Element	Decision
Unit of randomization	User , at the moment they enter the onboarding flow (feature-flag exposure $\approx$ signup start).
Split	50/50 , treatment (new onboarding) vs control (existing onboarding).
Eligibility	Every new signup entering onboarding during the window, on a supported app version. Existing users are not re-onboarded and are out of frame.
Assignment	Feature-flag service hashes a stable user id $\rightarrow$ arm. Assignment timestamp and the serving app version are logged.
Treatment	Fewer onboarding steps; identity-document upload deferred to after account creation (account opens limited; document gates limit increases and the physical card).
Control	Current onboarding: full KYC document upload before the account is created.
Primary metric	Day-7 Activation rate = share of randomized users with account created and ≥ 1 transaction within 7 days of assignment.
Guardrail metrics	(1) KYC rejection rate — share of submissions rejected; must not worsen (non-negotiable — fintech). (2) Flagged-fraud rate — accounts with a fraud signal in the first 30 days; must not worsen (non-negotiable). (3) Onboarding support tickets per 1,000 signups — operational-cost guardrail. Health metric: time-to-first-transaction.
Planned analyses	SRM / balance check (see §4.3 — here it is continuous , not pre-flight only); stated hypothesis + MDE / power for the 6-week window ; primary effect with 95% CI; heterogeneity by acquisition channel with a formal arm $\times$ channel interaction test; novelty check across the window.
Planning values (to finalize in Phase 1)	Control Day-7 Activation $\approx$ 48% ; target detectable lift $\approx$ +3 pp ; $\alpha = 0.05$ two-sided, power = 80%; $\sim 6,750/\text{arm} \rightarrow \text{MDE} \approx$ 2.4–2.6 pp .

4.2 The break (deterministic, baked in Phase 3)

The 2026-07-16 mobile deploy reset the on-device variant-assignment cache for a subset of users. Two distinct effects, which the investigation must **separate**:

1. **Contamination** — users **already assigned before the deploy** whose cache was reset (`cache_reset_flag = 1`). A fraction of them received a **second assignment** on their next app open, sometimes to the **other arm** $\rightarrow$ they experienced part of one arm's treatment and part of the other's, mid-journey.
2. **Post-deploy attribution bias** — **new signups after the deploy** fell into an assignment **fallback** (`assignment_source = 'fallback'`) that does **not** randomize correctly: the fallback arm is a deterministic function of **region / timezone**. So the post-deploy arms differ **in composition** (geography, and everything correlated with it), not only in **proportion**. The fallback share decays as the 2026-07-22 hotfix rolls out.

4.3 SRM monitoring as a standing process (not a one-off)

Unlike AeroVanti — where the balance check is a single **pre-flight** test done once before trusting the result — LumaBank runs SRM monitoring **continuously for the life of every experiment**. Documented in Phase 1 as an operational **runbook**: a daily job runs a chi-square test on cumulative and trailing-7-day arm counts; an alert fires at **$p < 0.001$ confirmed on two consecutive days**; escalation pauses enrollment and notifies the experiment owner and platform on-call; the triage checklist checks recent releases, the `assignment_source` mix, and the per-region split. **This monitoring is what catches the bug** — it is the difference between this project and AeroVanti.

4.4 The decision (three options, documented with trade-offs)

Option	Bias	Statistical power	Timeline	Note
1. Truncate — use only data up to the day before the deploy	Low (clean randomization pre-deploy)	Very low — ~ 10 days, $\sim 1/4$ of planned N, and assignments in days 4–10 are right-censored before their Day-7 window closes	Immediate, but inconclusive	Honest but almost certainly can't clear the MDE
2. Exclude only the contaminated — keep clean pre-deploy users; drop arm-switchers; drop or re-weight the biased post-deploy fallback	Residual — can't prove every contaminated user was found; re-weighting rests on the observed covariates only	Moderate — loses $\sim 30\text{--}40\%$ of the post-deploy sample	A few days of analysis, no new data	Fastest defensible answer, but reviewer-fragile
3. Restart clean — fix the cache/fallback bug, re-randomize, run	None	Recomputed for whatever window the business allows	Worst — but capped at 4	The choice

Option	Bias	Statistical power	Timeline	Note
again			weeks, not another 6	

Chosen: restart, in a **4-week** window with the bug fixed and **power/MDE recomputed from scratch** for the ~9,000 users available — *not* the original 6-week MDE reused. The deep dive states the explicit cost of the deadline: at ~4,500/arm the re-run can only reliably detect a lift of roughly ≥ 3 pp, so a true effect in the **2–3 pp** range — which the original 6-week design was powered for — would now likely be missed.

4.5 The final readout (re-run, clean)

Hypothesis restated; power recomputed for the 4-week window; **primary effect** (Day-7 Activation, treatment – control) with 95% CI; **guardrails** — KYC rejection rate and flagged-fraud rate treated as **non-negotiable** (a faster activation bought at the cost of letting risk through is a fail), plus support-ticket load; **heterogeneity by acquisition channel** (paid social, organic, referral, app-store search, influencer) with a formal interaction test.

Underlying true effect: decided during Phase 3 (ETL) and baked into the generated data with plausible noise and channel heterogeneity. The true values live only in `etl/config.py`; Phase 10 must detect and quantify them **as if blind**. The bug magnitudes (reset share, arm-switch fraction, fallback share and its region skew) are likewise set in Phase 3 and recovered by the investigation, not read off.

5. Data-quality issues to inject (generic set)

- 1. **Multiple date/time formats** across source exports — `DD/MM/YYYY` in the onboarding CSV, ISO-8601 in the KYC feed, Unix epoch-milliseconds in the transaction log.
- 2. **Timezone drift** — the risk/fraud system logs in **UTC**; everything else is `America/Sao_Paulo`, with no offset column.
- 3. **Money as strings** — `"R$ 1.234,56"`, `"1234.56"`, `"1.234,56"` mixed in transaction-amount and fee fields.
- 4. **Mojibake / mixed encoding** in user names and city fields.
- 5. **Duplicate rows** — the app retries on flaky mobile connections, so a slice of onboarding-step events and variant-assignment events are exact or near-exact duplicates.
- 6. **Missing master rows** — some `user_id`s appear in event feeds with no row in the user dimension export (referential gap).
- 7. **Snapshot-vs-log drift** — a "current variant" snapshot table that disagrees with the variant-assignment **event log**. (Diagnostically relevant: this disagreement is one of the first visible symptoms of the contamination.)

6. Bug-specific data catalog (the break, injected on purpose in Phase 3)

Element	Behaviour to simulate
<code>cache_reset_flag</code>	Emitted for ~12–18% of users already assigned before 2026-07-16 , on or shortly after the deploy.
Arm switch	~40% of the reset users get a second <code>variant_assignments</code> row on next app open; of those, ~50% land in the opposite arm → contaminated.
<code>assignment_source = 'fallback'</code>	For signups after 2026-07-16 : ~30–40% on day 1, decaying to ~0 over ~5 days as the 2026-07-22 hotfix rolls out.
Fallback bias	Fallback arm = deterministic hash of region / timezone , not the RNG → post-deploy arms diverge in geographic composition , not just in count.
<code>app_releases</code>	Contains the routine 2026-07-16 deploy (cause) and the 2026-07-22 hotfix (fix), among ~2–4 releases/month of noise.

7. Simulated source systems

System	Feeds	Grain
Feature-flag / experiment service	<code>variant_assignments</code> (user, arm, <code>assigned_ts</code> , <code>assignment_source</code> , <code>cache_reset_flag</code> , app version), <code>exposure_log</code> , <code>variant_snapshot</code> (current-state table)	assignment / exposure event
Onboarding / KYC service	<code>onboarding_steps</code> (step, ts), <code>kyc_submissions</code> (doc type, submitted_ts), <code>kyc_decisions</code> (approved / rejected / manual_review, reason code, decided_ts)	step / submission / decision

System	Feeds	Grain
Mobile app release log	<code>app_releases</code> (version, platform, <code>released_ts</code> , rollout %, notes) — the release calendar for root-cause cross-check	release
Core banking / ledger	<code>accounts</code> (user, opened_ts, status), <code>transactions</code> (account, ts, type, <code>amount_br1</code> , MCC) — first transaction = activation	account / transaction
Risk / fraud system	<code>fraud_signals</code> (user, flagged_ts UTC, signal type, score), <code>kyc_reason_codes</code> reference	fraud signal
CRM / marketing / support	<code>acquisition</code> (user, channel, campaign, first_touch_ts), <code>support_tickets</code> (user, opened_ts, category, channel)	user / ticket

8. Curated star schema (detailed in Phase 2)

- **Dimensions:** `dim_user`, `dim_date`, `dim_acquisition_channel` (paid social · organic · referral · app-store search · influencer), `dim_region` (UF + macro-region + timezone offset), `dim_experiment_arm` (control · treatment · ambiguous), `dim_app_release`, `dim_kyc_outcome`, `dim_onboarding_step`.
- **Facts:**
- `fact_variant_assignment` — **one row per assignment event** (contaminated users have ≥ 2): `user_key`, `arm_key`, `assigned_ts`, `assignment_source`, `cache_reset_flag`, `app_release_key`, `is_first_assignment`, `is_effective_assignment`.
- `fact_onboarding_step` — user × step × ts.
- `fact_kyc_decision` — one row per decision: `outcome_key`, `reason_code`, `is_rejected`.
- `fact_activation` — **one row per randomized user** (the analysis surface for the read-out): effective `arm_key`, `assigned_date_key`, `account_opened_flag`, `first_txn_ts`, `days_to_first_txn`, `day7_activated`, `cohort` (`pre_deploy` · `post_deploy` · `restart`), `contaminated_flag`, `in_analysis_flag`.
- `fact_transaction_day` — user × day (platform trend, time-to-first-transaction).
- `fact_fraud_event` — one row per fraud signal.
- `fact_support_ticket` — one row per ticket.
- `dq_checks.py` includes a check that **would catch the SRM as a data-quality failure** (arm-count chi-square by day), not only as an analysis step later.

9. Out of scope (explicit)

- A second country, region expansion, multi-currency / FX (covered by VoltEdge).
- Credit underwriting, limit-setting, lending — the account here is a checking account + card.
- A **fraud-detection model** — the flagged-fraud rate is a **guardrail input**, not modelled.
- Full customer-LTV / unit-economics modelling — activation is the outcome; downstream revenue is not modelled.
- Real-time / streaming data; everything is batch.
- App clickstream / screen-level telemetry beyond the onboarding-step events.
- Marketing-attribution modelling beyond a **first-touch channel** label.
- Real personal data — every user, account, transaction, KYC decision and fraud signal is synthetic and seed-generated.
- The **true treatment effect** and the **exact bug magnitudes** — set in Phase 3, detected blind in Phase 10.

10. Resolved-assumptions log

#	Assumption	Resolution
A1	Product	Free digital checking account + card. No credit/lending in frame.
A2	Window & anchor	21 months, 2025-01-01 → 2026-09-30; data-gen ("today") = 2026-09-10, so the last ~3 weeks are near-future relative to the build and exist only to close the re-run's Day-7 windows.
A3	Primary metric	Day-7 Activation = account created and ≥ 1 transaction within 7 days of assignment.
A4	"Active user"	≥ 1 transaction in the trailing 30 days (platform-health metric, not the experiment metric).
A5	Unit / split	User-level, 50/50, feature-flag hash assignment; assignment $\approx$ signup start.
A6	Original experiment window	6 weeks, 2026-07-06 → 2026-08-16; terminated early on the SRM alert.

#	Assumption	Resolution
A7	The break	Mobile deploy 2026-07-16 (day 11) resets the variant cache for a user subset; hotfix 2026-07-22.
A8	Two bug effects	(1) contamination of already-assigned users via arm switch; (2) region/timezone-correlated fallback bias for post-deploy signups. Baked in Phase 3, separated in Phase 10.
A9	SRM monitoring	Continuous daily chi-square (cumulative + trailing-7d); alert at $p < 0.001$ confirmed 2 consecutive days; documented as a runbook in Phase 1.
A10	Decision	Restart, not truncate or exclude-contaminated. Rationale + trade-off table in the deep dive.
A11	Re-run window	4 weeks, 2026-08-24 → 2026-09-20; ~9,000 users; power/MDE recomputed — original 6-week MDE not reused.
A12	Guardrails	KYC rejection rate + flagged-fraud rate = non-negotiable; onboarding support tickets / 1k signups = secondary.
A13	Heterogeneity	By acquisition channel, with a formal arm×channel interaction test.
A14	Planning values	Control Day-7 $\approx 48\%$; target lift $\approx +3$ pp; $\alpha = 0.05$, power = 80%. Finalized in Phase 1; true effect set in Phase 3.
A15	Currency	BRL only; no FX.
A16	True effect & bug magnitudes	Not decided here — set in ETL (Phase 3), detected blind in Phase 10.

11. Folder scaffold (created)

LumaBank-Activation-Experiment/		
docs/	00 scope · 01 business context & KPIs (+ SRM runbook) · 02 architecture · 03 data dict · 04 ETL · 05 model · 06 DAX · 07 visual identity · 09 report structure (DQ lives in data/quality/dq_report.md, not a separate doc) · walkthrough.md · PDF bundle	
etl/	config (one SEED, true effect + bug magnitudes) · 01 generate raw · 02 clean/stage · 03 build curated · dq_checks (incl. SRM check) · doc + DAX generators · run_pipeline · vocab/	
data/		
raw/	source-faithful messy exports	(git-ignored, regenerated)
staging/	typed stg_* tables	(git-ignored, regenerated)
curated/	Kimball star, CSV + Parquet	(git-ignored, regenerated)
quality/	dq_report.md	
powerbi/	semantic model as code (TMDL) · PBIR report · theme · build script	
assets/	brand.py · logo / page-background generators	
data_analysis/	sql/ python/ findings/ parity/ deliverables/ outputs/ ← PRIMARY (5-block investigation narrative: setup → SRM monitoring → root cause → decision analysis → re-run readout)	
portfolio/	5 gallery images only	
README.md		

End of Phase 0 deliverable. Next gate: Phase 1 — docs/01_business_context_kpis.md (business context, KPI definitions, stakeholder personas, and the SRM monitoring runbook as a documented operational process). Review before Phase 2.

LumaBank — Business Context & KPI Framework

Document status: Phase 1 deliverable · authored at project kickoff, before any analysis. **Data generation date:** 2026-09-10 · **Platform-history window:** 2025-01-01 → 2026-09-30 (21 months — LumaBank's life to date). **Embedded experiment:** onboarding redesign A/B test — original 2026-07-06 → 2026-08-16 (terminated on day 11), clean re-run 2026-08-24 → 2026-09-20. **Audience:** LumaBank's Growth, Product, and Risk & Compliance leadership.

This document frames the problem and defines the metrics. It deliberately does **not** state whether the new onboarding worked, or by how much — that is the job of the read-out (Phase 10). Everything here was knowable *before* the analysis. The true underlying treatment effect, and the exact magnitudes of the production bug that broke the first run, are decided during ETL (Phase 3), baked into the synthetic data with plausible noise and channel heterogeneity, and kept only in `etl/config.py` ; Phase 10 must detect and quantify them **as if blind**.

PT — resumo. A LumaBank é um neobank brasileiro fictício (conta digital + cartão), lançado em jan/2025; a janela de 21 meses é a vida inteira da plataforma. O problema: muita gente se cadastra e **nunca transaciona** — o fluxo de onboarding/KYC (upload de documento *antes* de a conta existir) derruba os usuários. Growth/Produto redesenhou o fluxo (menos etapas, documento adiado para *depois* da criação da conta) e rodou um **A/B test randomizado por usuário, 50/50**, com métrica primária **Day-7 Activation** (conta criada e primeira transação até o 7º dia). Este documento define contexto, KPIs, personas e o **runbook de monitoramento de SRM** — o processo operacional diário que, no dia 11 do experimento, detectou a quebra.

1. The situation

LumaBank is a Brazilian neobank. A user downloads the app, opens a free digital checking account and gets a Mastercard — in principle in minutes, entirely in-app. The top of the funnel is healthy: paid social, referral, app-store search and influencer campaigns all deliver signups at acceptable cost. The problem is what happens next.

A large share of new users never transact. They install the app, begin sign-up, reach the **KYC step** — a full identity-document upload (RG or CNH, plus a selfie/liveness check) that today happens *before* the account is created — and drop. Of those who do get through, many take more than a week to make a first payment or transfer, by which point the initial intent that brought them in has cooled and a competitor's card is already in their wallet.

The Growth and Product teams' hypothesis: **the document-upload wall is in the wrong place**. A user who has to photograph two documents and pass a liveness check *before they have anything* is being asked to pay a cost before receiving any value. Move the account creation ahead of the document upload — open the account in a **limited state** (low transaction and balance caps, no physical card) and request the documents *before those limits are lifted* — and more users will reach a first transaction, faster.

In LumaBank's 19th month of operation the teams ran this as a **randomized, user-level A/B test**. The rest of this project is what happened to that test: it **broke in production on day 11**, was caught by the standing SRM monitor (\$9), diagnosed, and re-run clean in a shorter window (\$10–11).

2. Company profile

Name	LumaBank (legal entity <i>LumaBank Instituição de Pagamento S.A.</i>)
Type	Neobank / <i>instituição de pagamento</i> — digital checking account (<i>conta de pagamento</i>) + Mastercard debit; no credit product in scope
Headquarters	São Paulo, SP, Brazil
Launched	January 2025 (public launch). This project's 21-month window is the platform's whole life to date.
Market / currency	Brazil only, BRL only. No FX, no cross-border — that ground is covered by the VoltEdge project.
Reporting calendar	Calendar months; growth reporting weekly; experiment reporting daily
Scale at window end	~ 130,000 cumulative signups · ~ 60,000 active users (≥ 1 transaction in the trailing 30 days) · end-of-window signup run-rate ~ 2,200–2,600 / week
Regulatory frame	Bacen <i>instituição de pagamento</i> rules; KYC/AML obligations (identity verification, sanctions/PEP screening, transaction monitoring). KYC and fraud controls are a hard constraint , not a dial to trade away for growth — this is why they are non-negotiable guardrails

	(\$8.3).

3. The product and the two onboarding flows

3.1 What a user gets

A free digital checking account, a virtual Mastercard immediately and a physical card on request, Pix (instant payments), boletos, transfers, and a simple in-app statement. Revenue (out of scope to model) is interchange on card spend, float, and Pix-for-business fees. The **activation** the business cares about is the user's **first transaction** — Pix, card purchase, boleto payment or transfer — because a user who transacts once is an order of magnitude more likely to still be here in 90 days.

3.2 Control flow ("old" — document upload before the account)

open app → phone + CPF + email → basic personal data → ****upload RG/CNH + selfie/liveness****
→ KYC decision (approve / manual review / reject) → account created → first transaction

The KYC document step sits **before** the account exists. A rejected or stuck KYC means the user never had an account at all.

3.3 Treatment flow ("new" — account first, documents deferred)

open app → phone + CPF + email → basic personal data → ****account created (LIMITED state)****
→ first transaction possible within limits → prompt to upload RG/CNH + selfie
→ KYC decision → limits lifted / physical card enabled

Fewer steps to a usable account; the document upload is deferred and gates only **limit increases and the physical card**, not the account itself. Limited-state caps (indicative, finalized in Phase 3): balance ≤ R\$ 1,000, Pix ≤ R\$ 500/day, no boleto issuance, no physical card.

3.4 Why this is a genuinely two-sided bet

Moving the KYC wall later is expected to lift activation — but it also means **unverified users can transact**, briefly and within caps, before the platform has confirmed who they are. That is exactly the population where **KYC rejection** and **first-party / synthetic fraud** concentrate. The experiment therefore cannot be read on the primary metric alone; the risk guardrails (\$8.3) decide as much as the activation lift does.

4. The activation funnel and its definitions

Stage	Definition
Signup started	User submits phone + CPF and enters the onboarding flow. This is the randomization point — feature-flag exposure is logged here with a timestamp and the serving app version.
KYC submitted	Identity documents + liveness uploaded (in control, before account creation; in treatment, after).
KYC decision	approved · manual_review (routed to an analyst queue, later resolved to approved/rejected) · rejected (with a reason code).
Account created	A usable account exists (in control, only after KYC approval; in treatment, at the limited-state step).
First transaction	First Pix, card purchase, boleto payment or transfer of any amount. The activation event.
Day-7 Activation (primary metric)	Account created and ≥1 transaction within 7 calendar days of the randomization timestamp . A user assigned on 2026-07-06 12:40 has until 2026-07-13 12:40.
Day-1 / Day-30 Activation	Same construct at 1-day and 30-day horizons (context metrics).
Time-to-first-transaction (TTFT)	Hours from randomization to first transaction, for activated users. Reported as median and p90.
Active user	≥1 transaction in the trailing 30 days. Platform-health metric — not the experiment metric.

Indicative control-flow funnel, per 100 signups started (planning values; the real numbers come from the pre-period in Phase 10):

Signup started	KYC submitted	KYC approved	Account created	Day-7 Activation	Day-30 Activation
100	~82	~74	~74	~48	~58

The ~18-point fall from "signup started" to "KYC submitted" — the document-upload wall — is the drop the redesign is aimed at.

5. Acquisition channels

Randomization is independent of channel, but channel is the pre-registered **heterogeneity dimension** (§8.6): the redesign may help low-intent, top-of-funnel traffic (which bounces hardest at the document wall) more than warm, high-intent traffic — or the reverse. This is a prior to **test**, not to assume.

Channel	~Share of signups	Indicative control Day-7 Activation	Character
Paid social (Meta, TikTok)	~38%	~42%	Highest volume, lowest intent, most price/creative-driven
Organic / direct	~22%	~54%	Searched for LumaBank by name; warm
App-store search (ASO)	~18%	~50%	Category intent ("conta digital"), medium-warm
Referral (member-get-member)	~15%	~58%	Invited by an existing user; warmest, but capped volume
Influencer / affiliate	~7%	~40%	Spiky with campaigns; intent varies wildly by creator

`dim_acquisition_channel` also carries campaign and a first-touch timestamp; only the **first-touch channel** is modelled (no multi-touch attribution — out of scope, §16).

6. KYC, risk and fraud (guardrail context)

The controls that the experiment must not damage:

Control	Definition	Indicative baseline (control flow)
KYC rejection rate	Share of KYC submissions with a final decision of <code>rejected</code> (document invalid/illegible, identity mismatch, sanctions/PEP hit, suspected forgery).	~9% of submissions
KYC manual-review rate	Share routed to an analyst queue before resolution — an operational-cost signal.	~4% of submissions
Flagged-fraud rate	Share of activated accounts with a confirmed fraud signal (first-party default intent, synthetic identity, account-takeover, mule activity) within the first 30 days .	~1.2% of activated accounts
Chargeback / dispute rate	Card disputes per 1,000 card transactions in the first 30 days.	context only

Fraud signals come from the risk system (rules + model score + confirmed-loss backfill) and are timestamped in **UTC** while every other system is `America/Sao_Paulo` — a deliberate data-quality issue (§ architecture doc). The risk system is a **source of a guardrail metric**; this project does not build or tune a fraud model (§16).

7. The analytics window — why 21 months

Anchor	Value
Window length	21 complete months: month 1 = Jan 2025 , month 21 = Sep 2026
Pre-experiment baseline	months 1–17 (Jan 2025 – May 2026) — the activation trend before any test
Original experiment	month 19 → 2026-07-06 → 2026-08-16 (6 weeks planned; stopped day 11)
Clean re-run	months 20–21 → 2026-08-24 → 2026-09-20 (4 weeks)
Outcome tail	to 2026-09-30 — needed to close the Day-7 window for re-run users assigned as late as 2026-09-20 (outcomes through 2026-09-27)

Why 21 months. It is LumaBank's entire operating history — there is no earlier comparable data. The window also gives a long, clean pre-period so the experiment's control arm can be sense-checked against the platform's established activation rate.

Near-future caveat. The build ("today") is **2026-09-10**; the last ~3 weeks of the window are dated after it and exist only to let the re-run's Day-7 outcomes complete. No analysis treats them as observed history.

8. The onboarding experiment

8.1 Design (as pre-registered, before the break)

Element	Decision
Unit of randomization	User , at signup start. The feature-flag service hashes a stable pre-account user id → arm; the assignment and the serving app version are logged.
Split	50/50 , treatment (new flow, \$3.3) vs control (old flow, \$3.2).
Eligible population	Every new signup entering onboarding during the window, on a supported app version ($\geq$ the version that ships the new flow behind the flag). Existing users are never re-onboarded and are out of frame.
Assignment integrity	Set once at signup start; no re-randomization; no mid-flight flow changes for either arm. (<i>This is precisely the invariant the production bug violates.</i>)
Window	6 weeks: 2026-07-06 → 2026-08-16 . Chosen partly because July is a lower-volume winter-holiday stretch and 6 weeks were needed to accrue the target sample.
Expected sample	~ 13,500 users (~2,250/week) → ~ 6,750 per arm .

8.2 Hypotheses

- **Business hypothesis.** Creating the account *before* the document upload, and deferring KYC to a limits-gate, raises the share of new users who complete a first transaction within 7 days — **without** worsening KYC rejection or fraud, and without a material rise in onboarding support load.
- **Statistical hypothesis (primary).** H_0 : Day-7 Activation is equal in the two arms. H_1 : treatment > control. Tested one-sided at $\alpha = 0.05$; a two-sided 95% CI on the difference is always reported.

8.3 Metrics

Role	Metric	Definition	Decision rule
Primary	Day-7 Activation rate	\$4. Share of assigned users with account created and ≥ 1 transaction within 7 days of assignment.	Ship if the lift is positive, significant ($\alpha = 0.05$) and $\geq$ the MDE .
Guardrail — non-negotiable	KYC rejection rate	Rejected ÷ KYC submissions, by arm.	Non-inferiority: treatment must not exceed control by more than 1.5 pp (95% one-sided). A breach fails the experiment regardless of the activation lift.
Guardrail — non-negotiable	Flagged-fraud rate (first 30 days)	Confirmed fraud signals ÷ activated accounts, by arm. Uses only the fraud window observable within the data.	Non-inferiority: treatment must not exceed control by more than 0.5 pp (95% one-sided). A breach fails the experiment.
Guardrail — secondary	Onboarding support tickets per 1,000 signups	Onboarding-category tickets ÷ signups × 1,000, by arm.	Directional; flag if treatment is materially worse (operational cost).
Health / context	TTFT (median, p90) , Day-1 & Day-30 Activation, limited-state-never-verified share (treatment only)	\$4.	Context for whether a Day-7 lift is durable and verified, not just deferred paperwork.

8.4 Power & minimum detectable effect — the 6-week design

Assumption	Value
Control Day-7 Activation (planning value)	48%
Target effect (MDE), absolute	+3 pp (48% → 51%)
α (two-sided) / power	0.05 / 0.80

Assumption	Value
Required n per arm at the MDE (two-proportion)	≈ 4,450
Expected n per arm	≈ 6,750 → power ≈ 0.93 at +3 pp; ~0.80 power for an effect as small as +2.4 pp
Per-channel cells (per arm, expected)	paid social ~2,570 · organic ~1,485 · ASO ~1,215 · referral ~1,010 · influencer ~475 → well powered for paid social/organic, directional for referral/influencer

The power calculation is re-run from realized pre-period rates and reproduced in code in Phase 10; these are the planning basis. **These numbers do not carry over to the re-run** — §11 recomputes them for the shorter window and smaller sample.

8.5 Threats to validity the read-out must address

Threat	How it is handled
Sample-ratio mismatch (SRM)	Continuously monitored , not checked once — see the runbook, §9. χ^2 goodness-of-fit on arm counts, daily, overall and by segment. <i>This is the monitor that caught the break.</i>
Assignment integrity failure	The variant snapshot table is reconciled against the assignment event log every day; disagreements (arm switches) are surfaced as a data-quality failure, not silently resolved.
Covariate imbalance	Standardized mean differences on channel, region, app version, OS, signup hour, device age; Love plot. Especially important after the break, where the fallback biases composition.
Novelty effect	Daily treatment Day-7 Activation across the window; test for a decaying trend; report the settled lift.
Seasonality	Concurrent control arm differences out calendar effects; both arms' daily series still reported.
Interference between arms	Referral links can put an inviter and invitee in different arms; assumed second-order, stated as a bounded limitation.
Multiple comparisons	One primary metric fixed in advance; the two hard guardrails are one-sided NI tests; the channel heterogeneity cut is secondary and CI-reported with a family-wise-error note.
Peeking	The integrity monitor runs daily by design; the effect analysis is a single fixed read at window close. Daily SRM looks never estimate the treatment effect.

8.6 Heterogeneity analysis (required)

Effect on Day-7 Activation estimated **within each acquisition channel**, with per-channel CIs and a formal **arm × channel interaction test** (likelihood-ratio). The business decision — full rollout vs rollout scoped to specific channels — depends on this cut.

9. Runbook — continuous SRM & assignment-integrity monitoring

This is a **standing operational process**, documented here because it is part of how LumaBank runs *every* experiment — not a one-off analysis step. It is what detected the break on 2026-07-17. It is owned by the Experimentation Platform team; the on-call rotation carries the pager.

9.1 Purpose

Detect, within one day, any failure of the randomization to deliver the intended allocation (SRM) or the intended **stable** per-user assignment (integrity), for the full life of a running experiment.

9.2 Cadence & inputs

- Runs **daily at 07:00 BRT** as a batch job, for every experiment in `running` state.
- Reads: the prior day's `variant_assignments` events, the cumulative assignment log, the `variant_snapshot` current-state table, `app_releases` , and a backend deploy/flag-config changelog.

9.3 Tests

#	Test	Grain
T1	Pearson χ^2 goodness-of-fit vs expected 50/50	cumulative assignments to date
T2	Pearson χ^2 vs 50/50	trailing-7-day assignments

#	Test	Grain
T3	χ^2 vs 50/50, broken down	by <code>assignment_source</code> (normal / fallback), region, timezone, app version, OS
T4	Snapshot-vs-log reconciliation: count users whose current snapshot arm $\neq$ their first logged assignment arm	per user, aggregated
T5	First-assignment vs effective-assignment drift: count users with ≥ 2 assignment events	per user, aggregated

9.4 Alert logic

- **WARN** — T1 or T2 $p < 0.01$ on a single day, or T4/T5 count above a small fixed threshold. Logged, digested to the experiment owner; no pause.
- **ALERT** — T1 $p < 0.001$, or T2 $p < 0.001$ on two consecutive days, or T4/T5 rising day-over-day. Thresholds are deliberately tighter than 0.05 because the test runs every day for weeks (a naïve 0.05 would false-alarm on most experiments).

9.5 On ALERT

1. **Auto-pause new enrollment** into the experiment (existing assignments keep serving).
2. **Page** the experiment owner and the Experimentation Platform on-call.
3. **Freeze** an analysis snapshot as of the last day that passed all tests ("last clean day").
4. Open an incident doc; begin triage (§9.6).

9.6 Triage checklist (in order)

1. **Release diff.** Any app release, backend deploy, or flag-config change in the 48 h before drift onset? Cross-reference `app_releases` and the deploy changelog. (*In this incident: the 2026-07-16 mobile release.*)
2. **Source split.** Break SRM by `assignment_source`. Is `fallback` non-zero or rising? When did the first fallback assignment appear?
3. **Segment split.** Is the imbalance concentrated by region / timezone / app version / OS? A geographic concentration points to a biased fallback, not random drift.
4. **Integrity.** Quantify T4/T5 — how many users have switched arms, and were they assigned before or after the onset?
5. **Population partition.** Count users assigned **before** onset (candidate-clean, but check for cache-reset), users assigned **after** onset (candidate-biased), and the overlap that experienced a reset.

9.7 Outputs the runbook hands to the decision

Contaminated-vs-clean user counts with an auditable rule; number of usable clean days and the sample they contain; whether the defect is **still active** or already fixed by a hotfix; the covariate-imbalance profile of the post-onset population. These feed directly into the three-option decision analysis (§ scope doc §4.4, and Phase 10 block 4).

10. The break (summary — full treatment in Phase 10)

On **2026-07-16** (experiment day 11) a routine mobile app release reset the on-device variant-assignment cache for a subset of users. The 07:00 run on **2026-07-17** raised a WARN on T2; **2026-07-18** confirmed it to an ALERT and enrollment auto-paused. Two distinct effects, which the investigation must separate:

1. **Contamination** — users **already assigned before 2026-07-16** whose cache was reset (`cache_reset_flag = 1`); a fraction were re-assigned on next app open, some to the **opposite arm**, so they experienced a mix of both flows mid-journey.
2. **Post-deploy attribution bias** — signups **after 2026-07-16** fell into an assignment **fallback** (`assignment_source = 'fallback'`) whose arm is a deterministic function of **region / timezone**, not random. The post-deploy arms differ in **composition**, not only in count. The fallback share decays as the **2026-07-22 hotfix** rolls out.

The magnitudes (reset share, arm-switch fraction, fallback share and its geographic skew) are set in Phase 3 and recovered by the analysis, not read off.

11. The re-run

Element	Decision
Preconditions	Cache-reset handling fixed (a reset no longer triggers re-assignment); the biased fallback removed (fallback, if ever hit, now uses the same hash as the primary path). Verified in staging before restart.

Element	Decision
Fresh randomization	All users re-hashed; no carry-over of arms from the aborted run; aborted-run users entering again are re-randomized and flagged.
Window	4 weeks: 2026-08-24 → 2026-09-20. The business could not wait another full 6 weeks; 4 weeks was the agreed cap.
Expected sample	~ 9,000 users (~4,500/arm) — a slightly higher late-window run-rate plus modest pent-up demand.
Power / MDE — recomputed	At ~4,500/arm, $\alpha = 0.05$, power = 0.80 → MDE $\approx$ 3.0 pp (vs 2.4 pp for the 6-week design).
Stated cost of the deadline	A true effect in the +2 to +3 pp range — a plausible real size for this change, and detectable under the original design — will now likely come back non-significant . The deep dive states this explicitly rather than treating a null as "no effect".
Guardrails & heterogeneity	Unchanged from \$8.3 / \$8.6, re-powered for the smaller sample (guardrail NI tests lose power too — noted).

12. Seasonality and patterns embedded in the data

Pattern	Where it shows
January "organize my finances" surge	Signup volume peaks; paid-social CAC falls
Carnaval (Feb)	Signups and support contacts dip for ~1 week
Feb–Mar back-to-school	Secondary signup lift
Black Friday / November	Card-acquisition marketing push → signup spike; card-transaction spike
December 13th-salary	Transaction volume and first-transaction rate rise; more Day-7 activations per signup
Jun–Jul winter holidays	Lower signup volume — the original experiment sits here, which is why 6 weeks were planned
Late Aug–Sep	Volume recovers — the re-run runs into a rising-volume stretch
Paid-social volume	Tracks quarter-end marketing pushes; creative refreshes cause week-to-week swings
App releases	~2–4 per platform per month; the 2026-07-16 release was routine and not flagged as risky at the time
Weekly cycle	Signups skew to evenings and Sun–Tue; first transactions cluster on the assignment day and the following weekend

13. Stakeholder personas → report pages

Four personas, one standing question each — those four questions are the Phase-8 report pages (the dashboard is the secondary deliverable, kept lean).

#	Persona	Standing question	Page
1	VP Growth / GM	Is new-user activation healthy and trending up, and did the onboarding redesign work well enough to roll out?	Executive Overview
2	Product Lead, Onboarding	Where exactly do new users drop between signup, KYC and first transaction — and how does the new flow reshape that funnel?	Onboarding & Activation Funnel
3	Experimentation / Data Science Lead	Can the experiment be trusted — allocation over time, SRM, contamination — and once it is clean, what is the effect?	Experiment Integrity Monitor
4	Head of Risk & Compliance	Did faster activation cost anything on KYC rejection or fraud? Are the guardrails intact?	Experiment Readout (guardrail section)

14. KPI framework

Benchmark ranges are **directional**, drawn from public Brazilian-fintech and neobank commentary and payments-industry notes, adapted to a young single-country neobank. They set targets; they are not claims about named competitors.

14.1 Platform health & scale

KPI	Definition	Benchmark	Target
Cumulative signups	Distinct users who ever started onboarding	—	on plan curve
Active users	≥ 1 transaction in trailing 30 days	—	≥ 60k by window end
Active rate %	Active ÷ (accounts created)	45%–65%	≥ 55%
New signups / week	Signups starting in the week	—	≥ plan
Account-creation rate %	Accounts created ÷ signups started	60%–80%	≥ 75%
MAU / signup cohort retention	Share of a weekly signup cohort active at week n	M1 ≥ 55%	context

14.2 Acquisition

KPI	Definition	Benchmark	Target
Signups by channel	Count by first-touch channel	—	mix on plan
Channel mix %	Channel share of signups	~38/22/18/15/7	context
Day-7 Activation by channel	Primary metric split by channel	40%–58%	context / heterogeneity input
Blended Day-7 Activation	All channels, all signups	45%–52%	≥ 50%

14.3 Onboarding funnel

KPI	Definition	Benchmark	Target
Step conversion	Share advancing signup → KYC submit → KYC decision → account → first txn	—	raise the KYC-submit step
KYC-submit rate %	KYC submitted ÷ signups started	70%–88%	≥ 85% (the redesign's job)
Account-creation rate %	§14.1	60%–80%	≥ 75%
Median time in onboarding	Signup start → account created (minutes)	4–15	≤ 8
Limited-state-never-verified % (<i>treatment only</i>)	Accounts still limited (no KYC) 14 days after creation	—	≤ 15%

14.4 Activation

KPI	Definition	Benchmark	Target
Day-7 Activation rate % (primary)	§4	40%–55%	≥ 52%
Day-1 / Day-30 Activation rate %	§4	—	context
Median / p90 TTFT	Hours from assignment to first transaction (activated users)	median 12–48 h	↓
First-transaction type mix	Pix / card / boleto / transfer share of first transactions	Pix-led	context
Activation by signup week	Day-7 Activation of each weekly cohort	—	trend up

14.5 KYC & risk — guardrails

KPI	Definition	Benchmark	Target
KYC rejection rate %	Rejected ÷ KYC submissions	5%–12%	≤ 10% and not worsened by the experiment
KYC manual-review rate %	Manual-review ÷ submissions	3%–8%	≤ 6%
Flagged-fraud rate % (30d)	Confirmed fraud signals ÷ activated accounts, first 30 days	0.5%–2.0%	≤ 1.5% and not worsened by the experiment
Chargeback rate (30d)	Card disputes per 1,000 first-30-day card transactions	—	context
Fraud-loss BRL per activated account	Confirmed loss ÷ activated accounts	—	context

14.6 Support & operating cost

KPI	Definition	Benchmark	Target
Onboarding tickets per 1,000 signups	Onboarding-category tickets ÷ signups × 1,000	60–120	≤ 85
Onboarding contact rate %	Signups with ≥1 onboarding ticket	4%–12%	≤ 8%
KYC-appeal rate %	Rejected users who contest the decision	—	context

14.7 Experiment scorecard (Integrity Monitor + Readout pages)

KPI	Definition
Day-7 Activation by arm (+ lift, 95% CI, p)	Primary metric, §8.3
Daily Day-7 Activation by arm	Novelty-effect diagnostic
Daily allocation by arm + cumulative SRM χ^2 p-value	Integrity monitor, §9 — the break story
SRM χ^2 by <code>assignment_source</code> / region / app version	Root-cause diagnostic
Snapshot-vs-log arm-switch count over time	Contamination diagnostic
Contaminated vs clean user counts	Auditable-rule output, §9.7
KYC rejection rate by arm (+ non-inferiority test)	Non-negotiable guardrail
Flagged-fraud rate by arm (+ non-inferiority test)	Non-negotiable guardrail
Onboarding tickets / 1k by arm	Secondary guardrail
Day-7 Activation effect by channel (+ arm × channel interaction test)	Heterogeneity, §8.6
Re-run power / MDE vs original	§11 — the cost of the deadline

15. Targets and plan

- The plan is built on the **same basis as the actuals** (same activation definition, same KYC-rejection and fraud-window definitions).
- **Plan Day-7 Activation:** rising to ≥ **52%** by window end (the redesign is the lever meant to get it there).
- **Plan active users:** ramp to **60k** by month 21.
- **Plan KYC-submit rate:** ≥ **85%**.
- **Plan KYC rejection rate:** hold ≤ **10%** — a ceiling, not a target to optimise down at the cost of legitimate approvals.
- **Plan flagged-fraud rate:** hold ≤ **1.5%**.
- **Plan onboarding tickets:** ≤ **85 / 1,000 signups**.
- "vs Target %" measures exist per KPI; **KYC rejection rate, flagged-fraud rate, manual- review rate, tickets per 1,000 and TTFT are scored lower-is-better**.

16. Out of scope (explicit)

- A second country, region expansion, multi-currency / FX (covered by VoltEdge).
- Credit / lending — underwriting, limits as a credit decision, interest income. The account here is a checking account + debit card; "limits" are a KYC gate, not credit.
- A **fraud-detection model** — the flagged-fraud rate is a guardrail **input**, produced by the existing risk system; this project does not build or tune it.
- Customer-LTV, contribution-margin or unit-economics modelling — activation is the outcome; downstream revenue (interchange, float) is not modelled.
- Marketing-attribution modelling beyond a **first-touch channel** label; no CAC/ROAS.
- App clickstream / screen-level telemetry beyond the defined onboarding-step events.
- Real-time / streaming data; everything is daily batch.
- Real personal data — every user, account, transaction, KYC decision and fraud signal is synthetic and seed-generated.
- The **true treatment effect** and the **exact production-bug magnitudes** — set in Phase 3, detected blind in Phase 10.

17. Resolved-assumptions log

#	Decision	Resolution
A1	Window length & anchor	21 months, Jan 2025 – Sep 2026; data-gen date 2026-09-10; = LumaBank's life to date.
A2	Near-future tail	The last ~3 weeks (after 2026-09-10) exist only to close the re-run's Day-7 windows; not treated as observed history.
A3	Product	Free digital checking account + Mastercard debit. No credit/lending in frame.
A4	Primary metric	Day-7 Activation = account created and ≥ 1 transaction within 7 days of the randomization timestamp.
A5	Randomization point	Signup start (feature-flag exposure), logged with timestamp + app version.
A6	"Active user"	≥ 1 transaction in the trailing 30 days (platform-health metric, not the experiment metric).
A7	Control vs treatment flow	Control = document upload before account creation; treatment = account created in limited state, KYC deferred to a limits gate.
A8	Unit / split	User-level, 50/50, feature-flag hash.
A9	Original experiment window	6 weeks, 2026-07-06 → 2026-08-16; terminated on day 11 by the SRM ALERT.
A10	Expected sample (original)	~13,500 users, ~6,750/arm.
A11	Guardrails	KYC rejection rate (NI, +1.5 pp bound) and flagged-fraud rate (NI, +0.5 pp bound) = non-negotiable; onboarding tickets / 1k = secondary.
A12	Power / MDE (original)	Control Day-7 $\approx$ 48%; MDE +3 pp; α 0.05, power 0.80; ~4,450/arm required, ~6,750 expected.
A13	Heterogeneity	By acquisition channel, with a formal arm $\times$ channel interaction test.
A14	SRM monitoring	Continuous daily χ^2 (cumulative + trailing-7d + segment splits) with WARN at $p < 0.01$, ALERT at $p < 0.001$; runbook in §9.
A15	The break	Mobile release 2026-07-16 resets the variant cache; two effects (contamination + region/timezone-biased fallback); hotfix 2026-07-22.
A16	The decision	Restart, not truncate or exclude-contaminated (trade-off table in scope §4.4 / Phase 10 block 4).
A17	Re-run window & power	4 weeks, 2026-08-24 → 2026-09-20; ~9,000 users; power/MDE recomputed → MDE $\approx$ 3.0 pp; deadline cost stated.
A18	Currency	BRL only; no FX in scope.
A19	True effect & bug magnitudes	Not decided here — set in ETL (Phase 3), detected blind in Phase 10.

End of Phase 1 deliverable. Next gate: Phase 2 — `docs/02_data_architecture.md` (medallion pipeline, the 6 simulated source systems, the curated star schema, and the data-quality + bug-specific injection catalog).

LumaBank — Data Architecture

Document status: Phase 2 deliverable. **Depends on:** `01_business_context_kpis.md` . **Feeds:** Phase 3 (ETL build), Phase 4 (data dictionary / model docs), Phase 6 (semantic model), **Phase 10 (analysis — primary deliverable)**.

This is a build spec: every layer, every table, every data-quality issue Phase 3 must inject and then handle — **plus** the bug-specific catalog (§9) that makes the first experiment run break on 2026-07-16. Scope note from Phase 0: the messy-data layer is **balanced** — 6 simulated source systems, 7 generic data-quality classes, and a 3-item bug catalog. Effort is concentrated on a curated layer that serves the incident narrative and the clean re-run read-out.

PT — resumo. Pipeline medallion (raw → staging → curated), tudo seedado e idempotente. 6 sistemas-fonte simulados. O star schema curated entrega uma linha por usuário randomizado (`fact_activation`) já com Day-7 Activation e todos os guardrails calculados, mais uma tabela `fact_srm_daily` (uma linha por dia de experimento) que alimenta o monitor de integridade. A quebra do dia 16/07 **não é ruído** — é injetada de propósito por `01_generate_raw.py` de forma determinística (eventos de cache-reset, troca de braço, e fallback de atribuição enviesado por região/timezone), e o `dq_checks.py` tem um check que **pegaria o SRM** como falha de qualidade de dados.

1. Principles — medallion, seeded, idempotent

Data only ever flows downward.

Layer	Folder	Contract
Raw	<code>data/raw/<source>/</code>	Byte-faithful simulated exports — messy, locale-specific, duplicated, encoded however the fake source system would encode it. Never edited after generation. Regenerated only when <code>SEED</code> / <code>config.py</code> changes.
Staging	<code>data/staging/</code>	Exactly one typed table per raw entity: <code>stg_<source>_<entity>.parquet</code> . Parsing, type coercion, encoding repair, de-duplication, key & category normalisation. No cross-source joins, no business logic, no metric definitions. Every fix counted into a DQ fragment.
Curated	<code>data/curated/</code>	Conformed Kimball star: <code>dim_*</code> / <code>fact_*</code> as Parquet and a CSV mirror. Surrogate keys, effective-arm resolution from the assignment event log, the auditable contamination rule, the pre-built <code>fact_activation</code> and <code>fact_srm_daily</code> tables. All money in BRL .
Quality	<code>data/quality/</code>	DQ fragments from staging + curated, assembled into <code>dq_report.md</code> ; hard-assertion results from <code>dq_checks.py</code> .

Seeded. A single `SEED` in `etl/config.py` drives every random draw. Same seed → byte-identical raw → identical curated. **The true treatment effect and the production-bug magnitudes live only in `config.py`** (see §10).

Idempotent. Each script wipes and rebuilds its own layer. `run_pipeline.py` runs `01 → 02 → 03 → dq_checks` end to end; `dq_checks.py` exits non-zero on any violated assertion.

Layered. Staging reads only raw. Curated reads only staging. The semantic model (Phase 6) and the analysis (Phase 10) read only `data/curated/*.parquet` .

2. Tech stack & run instructions

Concern	Choice
Language	Python 3.11+
Core libs	<code>pandas</code> , <code>numpy</code> , <code>pyarrow</code> (Parquet), <code>python-dateutil</code> , <code>openpyxl</code> (write messy <code>.xlsx</code>), <code>Faker</code> (<code>pt_BR</code> — names, cities, CPF-like ids), <code>ftfy</code> + <code>unicodedata</code> (encoding repair in staging), <code>duckdb</code> (Phase 10 SQL track), <code>scipy</code> / <code>statsmodels</code> (Phase 10 tests — not used in ETL)
Storage	Flat files. Raw = <code>.csv</code> / <code>.xlsx</code> / <code>.csv.gz</code> . Staging & curated = <code>.parquet</code> (+ curated <code>.csv</code> mirror). No database server.
Orchestration	Plain scripts, ordered by <code>run_pipeline.py</code> . No Airflow / dbt.
Determinism	One <code>SEED</code> ; every generator derives a named sub-seed from it. The 2026-07-16 break is derived from <code>SEED</code> + <code>user_id</code> hashes + a deterministic app-open model — reproducible, not random .

Concern	Choice
Windows / OneDrive	Write to a temp path in the same folder, then atomic <code>os.replace</code> ; a retry-on-lock wrapper (<code>utils.write_parquet</code> , ~4 attempts, backoff) absorbs intermittent WinError 5/32 locks.

```
# from repo root
python -m pip install -r etl/requirements.txt
python etl/run_pipeline.py           # full build: raw -> staging -> curated -> dq_checks
python etl/run_pipeline.py --from 02 # rebuild staging + curated only
python etl/gen_data_dictionary.py    # regenerate docs/03_data_dictionary.md from disk
```

Run order inside `run_pipeline.py` :

- 1. `01_generate_raw.py` — writes `data/raw/**`
- 2. `02_clean_stage.py` — writes `data/staging/stg_*.parquet` + `data/quality/dq_fragments/stg_*.json`
- 3. `03_build_curated.py` — writes `data/curated/{dim,fact}_*.parquet, csv` + curated DQ fragments
- 4. `dq_checks.py` — hard assertions; assembles `data/quality/dq_report.md` ; non-zero exit on failure

3. Naming conventions

Thing	Rule	Example
Raw file	<code><source>__<entity>__<period-or-grain>.<ext></code>	<code>lumencore__transactions__2026-07.csv</code>
Staging table	<code>stg_<source>_<entity></code>	<code>stg_flagfox_variant_assignments</code>
Dimension	<code>dim_<noun></code>	<code>dim_user</code>
Fact	<code>fact_<process>_<grain></code>	<code>fact_variant_assignment</code>
Surrogate key	<code><entity>_key</code> — integer, curated only	<code>user_key</code>
Business key	<code><entity>_id</code> — preserved from source	<code>user_id</code>
Date	<code><role>_date</code> (date) · <code><role>_date_key</code> (int YYYYMMDD)	<code>signup_date</code> , <code>signup_date_key</code>
Timestamp	<code><role>_ts</code> — tz-aware, user-local in curated (§7.3)	<code>assigned_ts</code> , <code>first_txn_ts</code>
Money	<code><measure>_brl</code> (single currency)	<code>txn_amount_brl</code>
Boolean	<code>is_<x></code> / <code>has_<x></code> / <code><x>_flag</code>	<code>is_rejection</code> , <code>day7_activated</code> , <code>contaminated_flag</code>
Columns	<code>snake_case</code> throughout curated	

Source timestamps are parsed to tz-aware UTC in staging, then curated derives the user's **local calendar day** (`America/Sao_Paulo` for all regions except the Amazonas / Acre timezones — §7.3) and every `*_date_key` from that local day, so day-boundary and experiment-day-boundary events land in the right bucket. **This matters for the Day-7 window and for the daily SRM counts.**

4. Source-system inventory

Six simulated systems. "Export style" is what makes each one messy.

#	System (fictional)	Owns	Raw entities	Export style / quirks
1	Lumen Core Banking	System of record: users, accounts, transactions, account-status events	<code>users_snapshot</code> (daily full dump), <code>accounts</code> , <code>transactions</code> (monthly), <code>account_status_events</code>	Daily full-dump user snapshot → massive row duplication ; <code>transactions</code> split into monthly files with overlapping date ranges → cross-file dupes; <code>amount</code> occasionally exported as <code>"R\$ 1.234,56"</code> ; <code>opened_at</code> mixes <code>YYYY-MM-DD</code> and <code>DD/MM/YYYY</code> ; transaction timestamp as Unix epoch-milliseconds .
2	Flagfox Feature-Flag & Experiment Service	A/B assignment for the onboarding test: arm, assignment source, cache-reset flag, exposure log,	<code>variant_assignments</code> , <code>exposure_log</code> , <code>variant_snapshot</code> (daily)	Modern tool, mostly clean — but <code>exposure_log</code> is at-least-once (duplicate rows); all timestamps UTC with Z while every other system is Brazil-local; from 2026-07-16 an <code>assignment_source = 'fallback'</code> value appears , a subset of rows carry <code>cache_reset = true</code> , and some users get a second <code>variant_assignments</code> row (the

#	System (fictional)	Owns	Raw entities	Export style / quirks
		current-variant snapshot		contamination); <code>variant_snapshot</code> drifts from the event log.
3	Trilha Onboarding Service	The signup funnel: step events, KYC submissions	<code>onboarding_step_events</code> , <code>kyc_submissions</code>	Step events at-least-once ; <code>step_name</code> free text (<code>kyc_upload</code> / <code>KYC_UPLOAD</code> / <code>doc_upload</code>); <code>occurred_at</code> ISO-8601 with offset; ~3% of sessions missing the terminal step (client dropped before the ack was written).
4	RiskGuard (KYC + Fraud)	KYC decisions & reason codes, fraud signals	<code>kyc_decisions</code> , <code>kyc_reason_codes</code> (ref), <code>fraud_signals</code>	Exports in UTC , no offset column; <code>decided_at</code> sometimes date-only ; reason codes as free text <i>and</i> code; fraud <code>signal_ts</code> as epoch-seconds ; ~1–2% of decisions reference a <code>user_id</code> absent from Core (queue lag).
5	Beacon Mobile-Platform Release Log	Mobile app releases + rollout; backend deploy / flag-config changelog	<code>app_releases</code> , <code>deploy_changelog</code>	<code>.xlsx</code> , one row per release; <code>released_at</code> as Excel serial ; <code>platform</code> free text (<code>ios</code> / <code>ios</code> / <code>Apple</code> , <code>Android</code> / <code>android</code>); <code>rollout_pct</code> as "85%" string. The routine 2026-07-16 mobile release and the 2026-07-22 hotfix are rows here , among ~2–4 releases/platform/month of noise.
6	Órbita Growth CRM & Support	First-touch acquisition channel + campaign; support tickets	<code>acquisition</code> , <code>support_tickets</code>	Free-text <code>channel</code> (<code>paid_social</code> / <code>Paid Social</code> / <code>meta</code>); <code>first_touch_at</code> mixed formats; CRM went live in month 3 , so months 1–2 signups have no acquisition row → <code>channel</code> unknown ; ticket <code>category</code> free text; <code>cidade</code> with Latin-1 encoding damage.

Flagfox (#2) corresponds to the experiment itself — a designed intervention, and the system the bug lives in. Beacon (#5) is the release calendar the root-cause investigation cross-references (doc 01 §9.6 step 1).

5. Raw layer — what `01_generate_raw.py` emits

Volumes are approximate at `SEED` default; calibrated in Phase 3 (§10 B3).

Raw file(s)	Grain	~Rows	Key messy / injected traits
<code>lumencore__users_snapshot__YYYY-MM-DD.csv</code> (daily)	user × snapshot day	~30M (dupe-heavy)	full re-dump daily; <code>opened_at</code> mixed date formats; <code>full_name</code> casing/whitespace noise
<code>lumencore__accounts.csv</code>	account	~100k	limited-state vs full-state; <code>status</code> free text; a few accounts with no matching user row
<code>lumencore__transactions__YYYY-MM.csv</code>	one transaction	~9M	overlapping monthly ranges → cross-file dups; <code>amount</code> sometimes "R\$..."; <code>ts</code> epoch-ms; <code>mcc</code> / <code>type</code> free text (<code>pix</code> / <code>PIX</code> / <code>transfer_pix</code>)
<code>lumencore__account_status_events.csv</code>	status-change event	~180k	duplicate <code>event_id</code> (at-least-once); <code>changed_at</code> ISO with offset
<code>flagfox__variant_assignments.csv</code>	assignment event	~24k	one row/user pre-2026-07-16; from 2026-07-16 : <code>assignment_source</code> ∈ {primary, fallback} , <code>cache_reset</code> ∈ {true,false} , and ~3% of pre-deploy users get a 2nd row with a different arm (BUG02); stratum - free (user-level)
<code>flagfox__exposure_log.csv</code>	user × exposure event	~600k	at-least-once duplicates; UTC <code>Z</code> timestamps; carries the <code>cache_reset</code> marker on the first post-deploy open for affected users (BUG01)
<code>flagfox__variant_snapshot__YYYY-MM-DD.csv</code> (daily)	user × snapshot day	~5M	"current arm" per user; drifts from the event log once contamination starts (Q07)
<code>trilha__onboarding_step_events.csv</code>	user × step event	~700k	at-least-once; <code>step_name</code> free text; missing terminal step for ~3%
<code>trilha__kyc_submissions.csv</code>	KYC submission	~110k	<code>submitted_at</code> ISO; <code>doc_type</code> free text (<code>rg</code> / <code>RG</code> / <code>cnh</code>); in treatment, submitted after account creation

Raw file(s)	Grain	~Rows	Key messy / injected traits
riskguard_kyc_decisions.csv	KYC decision	~110k	UTC , no offset; <code>decided_at</code> sometimes date-only; reason free text + code; ~1–2% orphan <code>user_id</code>
riskguard_kyc_reason_codes.csv	reason code	~30	reference; free-text labels
riskguard_fraud_signals.csv	fraud signal	~3k	<code>signal_ts</code> epoch-seconds (UTC); <code>signal_type</code> free text; <code>confirmed</code> flag backfilled
beacon_app_releases.xlsx	release	~150	Excel-serial <code>released_at</code> ; <code>platform</code> free text; <code>rollout_pct</code> as "NN%"; rows for the 2026-07-16 release + 2026-07-22 hotfix
beacon_deploy_changelog.csv	backend deploy / config change	~400	free-text <code>component</code> ; ISO timestamps
orbita_acquisition.csv	user	~128k	free-text <code>channel</code> ; mixed date formats; no rows for months 1–2
orbita_support_tickets.csv	ticket	~140k	free-text <code>category</code> / <code>channel</code> ; <code>cidade</code> mojibake; duplicate <code>ticket_id</code> (at-least-once)

6. Staging layer — 02_clean_stage.py

One `stg_*` table per raw entity. Responsibilities and the DQ counter each raises:

Fix class	What happens	DQ counter(s)
Date / time parsing	Parse <code>YYYY-MM-DD</code> , <code>DD/MM/YYYY</code> , ISO-8601 with offset, Excel serials (epoch 1899-12-30), epoch-ms (Core transactions) and epoch-s (fraud signals). Date-only <code>decided_at</code> → midnight UTC + <code>time_imputed</code> flag.	<code>dates_parsed</code> , <code>excel_serials_converted</code> , <code>epochs_converted</code> , <code>dates_unparseable</code>
Money parsing	Strip <code>R\$</code> , thousands <code>.</code> , and decimal <code>,</code> ; blank → null (never 0).	<code>money_strings_parsed</code>
Encoding repair	<code>ftfy.fix_text</code> + targeted Latin-1→UTF-8 re-decode for Órbita names / <code>cidade</code> ; strip BOM.	<code>mojibake_repaired</code> , <code>bom_stripped</code>
De-duplication	Drop exact-duplicate <code>event_id</code> / <code>ticket_id</code> / <code>exposure</code> / <code>step</code> rows; collapse the daily user full-dump to one row per user per day; collapse the daily <code>variant_snapshot</code> ; dedupe overlapping <code>transactions</code> monthly files on <code>txn_id</code> . variant_assignments rows are NOT deduped away — multiple rows per user are real signal (contamination) and are kept, ordered, with an <code>assignment_seq</code> .	<code>dupe_events_dropped</code> , <code>snapshot_rows_collapsed</code> , <code>dupe_txn_dropped</code>
Timezone	All timestamps → tz-aware UTC (Flagfox / RiskGuard already <code>Z</code> or UTC; others localised from their offset). Curated later derives user-local day.	<code>timestamps_localized</code>
Category harmonisation	Map <code>step_name</code> , transaction <code>type</code> , ticket <code>category</code> / <code>channel</code> , <code>channel</code> (acquisition), <code>platform</code> , KYC <code>outcome</code> / reason group, <code>doc_type</code> to controlled vocab in <code>etl/vocab/</code> . Unknowns → explicit "unknown" + logged.	<code>categories_normalised</code> , <code>category_unknowns</code>
Key normalisation	Trim business keys; cast to string; null-out placeholders (<code>"0"</code> , <code>" "</code> , <code>"N/A"</code>).	<code>null_keys_flagged</code>
Boolean coercion	<code>cache_reset</code> , <code>confirmed</code> , <code>is_award</code> etc. from <code>"true"/"1"/"S"/"sim"</code> → real bool.	<code>bools_coerced</code>
Birth-year sanity	<code>data_nascimento</code> ∈ {1900-01-01, 0, future} → null + <code>birth_year_valid</code> = false.	<code>birthyear_invalidated</code>
Type coercion	Enforce a declared dtype per column; failures → null + counter.	<code>type_coercion_failures</code>

Staging does **not** join across sources, does **not** compute activation / SRM / the effective arm, and does **not** drop rows for referential problems — that is curated's job, fully accounted.

7. Curated layer — 03_build_curated.py

Kimball star. `dim_date` is built first and spans **2024-06-01** → **2027-12-31** (past the fact window, so no event falls on a missing date row).

7.1 Dimensions

Table	Grain	Notable attributes
<code>dim_date</code>	calendar day	<code>date_key</code> , <code>y/q/m/iso-week</code> , <code>month_name</code> , <code>is_month_end</code> , <code>is_holiday_br</code> , <code>season_tag</code> , <code>experiment_phase</code> (pre / original / washout / rerun / post), <code>experiment_day</code> (1..N within original and within rerun, else null), <code>is_post_deploy</code> (original run, ≥ 2026-07-16), hidden sort keys
<code>dim_user</code>	user	<code>user_key</code> , <code>user_id</code> , <code>signup_date_key</code> , <code>signup_cohort_week</code> , <code>acquisition_channel_key</code> , <code>campaign</code> , <code>region_key</code> , <code>device_os</code> , <code>app_version_at_signup</code> , <code>age_band</code> , <code>birth_year_valid</code> , <code>is_test_account</code> , <code>experiment_arm</code> (control / treatment / ambiguous / not_in_experiment), <code>experiment_cohort</code> (pre_deploy / post_deploy / rerun / not_in_experiment), <code>is_experiment_subject</code> , <code>contaminated_flag</code> , <code>in_analysis_flag</code>
<code>dim_acquisition_channel</code>	channel	<code>channel_key</code> , <code>channel_name</code> (paid_social / organic / app_store_search / referral / influencer / unknown), <code>channel_group</code> (paid / owned / earned / unknown), <code>is_paid</code>
<code>dim_region</code>	UF	<code>region_key</code> , <code>uf</code> (27), <code>macro_region</code> (N / NE / CO / SE / S), <code>timezone</code> (America/Sao_Paulo, America/Manaus, America/Rio_Branco, ...), <code>utc_offset_hours</code> , <code>is_amazon_tz</code> — the axis the post-deploy fallback bias correlates with (BUG03)
<code>dim_app_release</code>	release	<code>app_release_key</code> , <code>version</code> , <code>platform</code> (iOS / Android / backend), <code>released_date_key</code> , <code>released_ts</code> , <code>rollout_pct</code> , <code>release_type</code> (feature / hotfix / config), <code>is_cache_reset_release</code> (true only for the 2026-07-16 row), <code>notes</code>
<code>dim_kyc_outcome</code>	outcome	<code>kyc_outcome_key</code> , <code>outcome</code> (approved / manual_review / rejected / pending), <code>reason_code</code> , <code>reason_group</code> (doc_quality / identity_mismatch / sanctions_pep / suspected_fraud / abandoned / n/a), <code>is_rejection</code>
<code>dim_onboarding_step</code>	step	<code>step_key</code> , <code>step_name</code> (signup_started / personal_data / kyc_upload / account_created / first_transaction / limits_lifted), <code>step_order</code> , <code>funnel_stage</code>
<code>dim_experiment_arm</code> (disconnected slicer helper, built Phase 6)	arm	<code>control</code> / <code>treatment</code> / <code>ambiguous</code>

7.2 Facts

Table	Grain	Measures / columns	Build notes
<code>fact_variant_assignment</code>	one assignment event (contaminated users have ≥2)	<code>user_key</code> , <code>arm_key</code> , <code>assigned_ts</code> , <code>assigned_date_key</code> , <code>assignment_seq</code> , <code>assignment_source</code> (primary / fallback), <code>cache_reset_flag</code> , <code>app_release_key</code> (release in effect at assignment), <code>is_first_assignment</code> , <code>is_effective_assignment</code>	Ordered by <code>assigned_ts</code> per user. <code>is_effective_assignment</code> set by the auditable rule in §7.3. Spine of the SRM / integrity analysis.
<code>fact_onboarding_step</code>	user × step (first occurrence)	<code>user_key</code> , <code>step_key</code> , <code>occurred_ts</code> , <code>occurred_date_key</code> , <code>seconds_since_signup</code> , <code>is_reached</code>	Deduped at-least-once; a step never reached → <code>is_reached</code> = false, no row for later steps. Funnel spine.
<code>fact_kyc_decision</code>	one KYC decision	<code>user_key</code> , <code>kyc_outcome_key</code> , <code>submitted_ts</code> , <code>decided_ts</code> , <code>decision_latency_hours</code> , <code>is_rejection</code> , <code>is_manual_review</code> , <code>doc_type</code> , <code>submitted_after_account</code>	<code>submitted_after_account</code> = true is the treatment-flow signature. Orphan <code>user_id</code> → sentinel key, counted.
<code>fact_activation</code>	one row per signup — platform-wide (<code>is_experiment_subject</code> flags the randomized ones)	<code>user_key</code> , <code>arm_key</code> (effective ; <code>not_in_experiment</code> outside the two runs), <code>assigned_ts</code> , <code>assigned_date_key</code> , <code>experiment_cohort</code> , <code>acquisition_channel_key</code> , <code>region_key</code> , <code>app_version_at_signup</code> ,	One tidy row per signup with every activation and guardrail metric pre-computed — both the Executive Overview's platform-wide activation trend and Phase 10's experiment read-out (filtered to <code>is_experiment_subject</code>) read this one table. <code>day7_activated</code>

Table	Grain	Measures / columns	Build notes
		account_opened_flag , account_opened_ts , first_txn_ts , hours_to_first_txn , day7_activated , day1_activated , day30_activated , kyc_submitted_flag , kyc_rejected_flag , fraud_flag_30d , onboarding_tickets_n , is_experiment_subject , contaminated_flag , assignment_switch_flag , in_analysis_flag	recomputed independently in dq_checks .
fact_transaction_day	user × day	txn_count , txn_amount_brl , is_first_txn_day , pix_count , card_count , boleto_count , transfer_count	Platform trend, TTFT, activity context. Deduped, epoch-ms parsed, user-local day.
fact_fraud_event	one fraud signal	user_key , flagged_ts (→ user-local), flagged_date_key , signal_type , signal_score , is_confirmed , days_since_activation	Feeds the fraud guardrail. Epoch-s + UTC converted.
fact_support_ticket	one ticket	user_key , opened_ts , opened_date_key , category (onboarding / kyc / card / transaction / other), channel (chat / email / phone), is_onboarding	Feeds the support-cost guardrail.
fact_srm_daily	experiment × calendar day	experiment_phase , assign_date_key , n_control , n_treatment , n_fallback , n_primary , srm_chisq_cumulative , srm_p_cumulative , srm_chisq_trailing7 , srm_p_trailing7 , snapshot_log_mismatch_n , multi_assignment_n , alert_state (ok / warn / alert)	The monitoring surface — drives the Integrity Monitor page and analysis block 2. Computed deterministically from fact_variant_assignment + variant_snapshot ; Phase 10 re-derives it independently (SQL + pandas) for parity. The 2026-07-16 break is present here by construction.
fact_targets_month	month × kpi	target_value	Plan from doc 01 §15; same basis as actuals.

7.3 Cross-cutting curated rules

- **Effective arm from the event log (auditable).** fact_activation.arm_key and fact_variant_assignment.is_effective_assignment follow one rule, stated here so Phase 10 is reproducible: a user's effective arm is their **first** assignment, **unless** a later assignment with a *different* arm has assigned_ts before **50 % of that user's Day-7 window has elapsed** — then the later arm is effective. Either way, a user with ≥2 differing-arm assignments is assignment_switch_flag = 1 .
- **Contamination rule (auditable).** contaminated_flag = 1 iff (a) the user has ≥2 assignment events with differing arms, **or** (b) the user has a cache_reset_flag event whose timestamp lies inside their active Day-7 window. This is the exact rule the "clean vs contaminated" split in Phase 10 block 3 uses.
- **experiment_cohort** . pre_deploy = original-run user assigned **before** 2026-07-16; post_deploy = original-run user assigned **on/after** 2026-07-16; rerun = assigned in the 2026-08-24 → 2026-09-20 window; else not_in_experiment . assignment_source and region_key are retained on every subject so the post-deploy composition bias is quantifiable.
- **in_analysis_flag — the clean re-run population.** = 1 iff experiment_cohort = 'rerun' **and** contaminated_flag = 0 **and** every assignment for the user has assignment_source = 'primary' . The **primary read-out filters on this**; the pre_deploy / post_deploy cohorts are analysed only for the incident narrative (blocks 2–4).
- **variant_snapshot is measurement-only.** Used solely to compute snapshot_log_mismatch_n (users whose snapshot arm ≠ their first logged arm) — never an input to arm_key .
- **Day-7 window from the user-local calendar.** hours_to_first_txn , day7_activated , day1/day30_activated computed from tz-aware timestamps; the local day uses the user's dim_region.timezone (America/Sao_Paulo for ~93 % of users;

America/Manaus / America/Rio_Branco for the Amazon-tz UFs). SRM daily counts bucket assignments by the **user-local** assigned_date_key for the same reason.

- **Timezone normalisation.** Flagfox + RiskGuard timestamps (UTC) → tz-aware → user-local. Fraud flagged_ts bucketed by user-local day.
- **Sentinel keys.** user_key = -1 (event references a user id absent from Core), user_key = -2 (KYC / fraud decision, unknown user). Facts are **never dropped** for a missing dimension; the sentinel is used and counted.
- **Experiment denormalisation.** experiment_arm / experiment_cohort / is_experiment_subject are stamped on dim_user and propagated to fact_transaction_day, fact_support_ticket, fact_fraud_event via user_key, so the report can slice platform metrics by arm without touching the experiment facts.
- **BRL only.** Every money column is *_brl. No _usd sibling, no FX table.

8. Data-quality issue catalogue — generic classes

Seven **generic** classes — injected by 01, handled by 02 / 03, counted into a DQ fragment, and (where an invariant) asserted in dq_checks.py.

#	Issue	Injected in	Handled in	Hard check
Q01	Multiple date/time formats + Excel serials + epoch (ms & s)	Core opened_at & transaction ts, Trilha ISO, RiskGuard date-only UTC, Beacon Excel serial, fraud epoch-s, Órbita mixed	staging date/epoch parser (locale + format rule per source)	dates_unparseable = 0
Q02	Money as locale strings ("R\$ 1.234,56", symbol-in-value)	Core transactions.amount, fee fields	staging money parser	no non-null money column is object dtype; txn_amount_br1 finite
Q03	Timezone drift — Flagfox & RiskGuard in UTC Z with no offset column, everything else BR-local; plus genuine multi-tz UFs (AM / AC / RO / RR)	Flagfox, RiskGuard, dim_region	staging → UTC; curated → user-local day	no naive timestamps in curated; SRM daily counts reconcile to raw within ±0 rows on a spot-checked day
Q04	Encoding damage (Latin-1 mojibake, BOM)	Órbita names / cidade, some full_name	ftfy + targeted re-decode	no Ã / Ä mojibake sequences remain in dim_user -joined city text
Q05	Duplicate rows — daily user full-dump, daily variant snapshot, overlapping monthly transactions, at-least-once exposure / step / status / ticket events	Core, Flagfox, Trilha, Órbita	staging de-dup (collapse / dedupe on id) — but variant_assignments multi-rows are kept as signal	fact_transaction_day reconciles to a unique- txn_id count; snapshot collapsed to 1 row / user / day; fact_variant_assignment unique on (user_id, assignment_seq)
Q06	Missing master rows / orphan keys — KYC / fraud decisions referencing users absent from Core (queue lag); months-1-2 signups with no Órbita acquisition row	RiskGuard, Órbita late go-live	curated sentinel keys + count; missing channel → unknown	every fact FK resolves to a real or sentinel key; orphan + unknown - channel counts reported
Q07	Snapshot-vs-log drift — Flagfox variant_snapshot disagrees with the variant_assignments event log	Flagfox	curated builds the effective arm from the event log; drift measured into snapshot_log_mismatch_n	curated arm_key derived from the log by construction ; drift-row count reported and is ~0 before 2026-07-16, then rises (a symptom, not a failure)

Also handled without ceremony (one counter each): null/placeholder keys → sentinels; invalid birth years → null + flag; negative / out-of-range amounts → clamped + counted; free-text step_name / type / category / channel / platform → controlled vocab; missing terminal onboarding step → is_reached = false.

9. Bug-specific catalogue — the break (injected deterministically)

Not noise. 01_generate_raw.py injects a **deterministic** production incident on 2026-07-16, driven by SEED + user_id hashes + an app-open timing model. All magnitudes are config.py parameters (\$10 B2), recovered — not read off — in Phase 10.

#	Mechanism	Behaviour to generate	Params
BUG01	Cache-reset events	The 2026-07-16 mobile release (<code>is_cache_reset_release</code>) emits <code>cache_reset = true</code> on the next <code>exposure_log</code> (and any new <code>variant_assignments</code>) row for a share of users assigned before 2026-07-16 who open the app within the following ~5 days.	<code>RESET_SHARE ≈ 0.15</code> , <code>RESET_OPEN_WINDOW_DAYS ≈ 5</code>
BUG02	Arm switch (contamination)	Of the reset users, a fraction get a fresh <code>variant_assignments</code> row on next open; its arm is a re-hash with a <i>different salt</i> , so $\approx 50\%$ land in the opposite arm. Net $\approx 3\%$ of the pre-deploy population becomes <code>contaminated_flag = 1</code> with <code>assignment_switch_flag = 1</code> .	<code>RESWITCH_SHARE ≈ 0.40</code>
BUG03	Biased post-deploy fallback	For signups on/after 2026-07-16 , a share — starting $\sim 35\%$ on 2026-07-16 and decaying to ~ 0 over ~ 5 days as the 2026-07-22 hotfix rolls — get <code>assignment_source = 'fallback'</code> . The fallback arm is <code>hash(uf timezone) mod 2</code> , <i>not</i> the RNG → not 50/50 and correlated with geography: SE/S UFs skew $\sim 58\%$ treatment, Amazon-tz UFs skew $\sim 65\%$ control. Produces SRM and covariate imbalance.	<code>FALLBACK_SHARE_0 ≈ 0.35</code> , <code>FALLBACK_DECAY_DAYS ≈ 5</code> , <code>FALLBACK_GEO_SKEW ≈ 0.58 / 0.65</code>
—	Hotfix	A <code>release_type = 'hotfix'</code> row on 2026-07-22 in <code>beacon_app_releases</code> ; after it, no new <code>fallback</code> or <code>cache_reset</code> rows. The re-run (2026-08-24+) is generated entirely on the clean path.	<code>HOTFIX_DATE = 2026-07-22</code>

`dq_checks.py` — hard assertions

- **Referential:** every fact FK resolves to a real or sentinel key.
- **Assignment integrity:** `fact_variant_assignment` unique on (`user_id` , `assignment_seq`); exactly one `is_first_assignment` per user; every `fact_activation.user_key` has ≥ 1 assignment row.
- **Activation identity:** `day7_activated == (account_opened_flag AND first_txn_ts ≤ assigned_ts + 7 days)` , recomputed independently from `fact_transaction_day` .
- **Funnel monotonicity:** `fact_onboarding_step` reached-counts are non-increasing across `step_order` within each flow.
- **The SRM check that would catch the bug:** `fact_srm_daily` , re-derived from `fact_variant_assignment` , must match the stored table within float tolerance; and it must show `srm_p_cumulative < 0.001` **somewhere inside the original run on/after 2026-07-17** and `srm_p_trailing7 ≥ 0.05` **for every day of the re-run**. The break is present; the re-run is clean.
- **Re-run cleanliness:** for all `in_analysis_flag = 1` users, `assignment_source = 'primary'` and `contaminated_flag = 0` (100 %).
- **Benchmark bands:** Day-7 Activation, KYC rejection rate, flagged-fraud rate and onboarding tickets / 1k land inside their doc 01 §14 ranges for the **pre-experiment** period. The re-run arms are allowed to diverge (that is the point); the original run is *expected* to look broken.
- **Near-future guard:** no `fact_transaction_day` rows after 2026-09-30; every `rerun` user assigned $\leq$ 2026-09-20 has a complete Day-7 window inside the data.

10. Architecture diagram

```
flowchart TD
    subgraph SRC["6 simulated source systems"]
        S1["Lumen Core Banking<br/>users - accounts - transactions - status events"]
        S2["Flagfox Feature-Flag & Experiment Service<br/>variant assignments - exposure log - variant snapshot"]
        S3["Trilha Onboarding Service<br/>step events - KYC submissions"]
        S4["RiskGuard (KYC + Fraud)<br/>KYC decisions - reason codes - fraud signals"]
        S5["Beacon Mobile-Platform Release Log<br/>app releases - rollout - deploy changelog"]
        S6["Orbita Growth CRM & Support<br/>first-touch channel - campaigns - support tickets"]
    end

    SRC -->| "01_generate_raw.py<br/>seeded, messy exports<br/>+ the deterministic 2026-07-16 break" | RAW["data/raw/**<br/>RAW -->| "02_clean_stage.py<br/>parse - repair - de-dup - normalise<br/>(keep multi-assignment rows as signal)<br/>STG -->| "03_build_curated.py<br/>conform - surrogate keys<br/>effective-arm rule - contamination rule<br/>fact_act"]
    STG --> DQF["data/quality/dq_fragments/"]
    CUR --> DQF
    DQF -->| "dq_checks.py<br/>hard assertions + the SRM check that catches the bug" | DQR["data/quality/dq_report.md<br/>DQR -->| "Phase 6 - gen_semantic_model.py" | SM["Power BI .SemanticModel<br/>TMDL + Parquet import partitions"]]
```

```
SM -->|"Phase 8 - build_report.sh (lite)"| RPT["Power BI .Report<br/>4 pages"]
CUR ==>|"Phase 10 - DuckDB (SQL) + pandas (Python)<br/>PRIMARY DELIVERABLE"| ANA["data_analysis/<br/>incident narr
```

11. Open items carried into Phase 3

#	Item	Resolution plan
B1	The true treatment effect — overall Day-7 Activation lift, per-channel heterogeneity profile, novelty-decay shape, and the guardrail impacts (KYC rejection rate, flagged-fraud rate, support tickets)	Decided in Phase 3 , encoded in <code>config.py</code> as tunable effect sizes + noise; kept out of docs 00 / 01; detected blind in Phase 10. Design intent: a real overall lift close to the re-run's ~3 pp MDE , so the "cost of the deadline" genuinely bites; effect larger for low-intent paid-social / influencer , smaller for referral / organic; a small early novelty bump; guardrails roughly neutral but with at least one (KYC rejection or fraud) drifting up enough to need a careful non-inferiority read .
B2	The production-bug magnitudes — <code>RESET_SHARE</code> , <code>RESET_OPEN_WINDOW_DAYS</code> , <code>RESWITCH_SHARE</code> , <code>FALLBACK_SHARE_0</code> , <code>FALLBACK_DECAY_DAYS</code> , <code>FALLBACK_GEO_SKEW</code> , <code>HOTFIX_DATE</code>	Set in <code>config.py</code> (§9). Chosen so the cumulative SRM p-value crosses 0.001 on ~day 12–13 and the trailing-7d test confirms it two days running — matching the doc 01 §9 runbook thresholds.
B3	Final row volumes vs the ~130k user / ~9M transaction targets	Calibrate <code>SEED</code> -default population and activity rates in the first regeneration pass.
B4	Benchmark-band calibration (Day-7 Activation, KYC rejection, fraud flag, tickets/1k, funnel step conversion)	Several regeneration passes until the pre-experiment period sits mid-band for every doc 01 §14 KPI.
B5	BR holiday / school-break calendar & fintech seasonality curve	Static lookup in <code>etl/vocab/br_calendar.csv</code> ; seasonal multipliers in <code>config.py</code> .
B6	UF → macro-region → timezone lookup; acquisition-channel & step vocab	<code>etl/vocab/br_uf_region_tz.csv</code> , <code>etl/vocab/channel_map.csv</code> , <code>etl/vocab/step_map.csv</code> .
B7	Exact experiment sample sizes	Generator targets ~2,250 signups/week entering the experiment → original run ~13,500 over 6 weeks (stopped at day 11 → ~2,600 usable pre-deploy + a contaminated/biased tail), re-run ~9,000 over 4 weeks.

End of Phase 2 deliverable. Next gate: Phase 3 — `etL/` (seeded generator with the deterministic break, staging, curated star, `dq_checks.py`), then `data/quality/dq_report.md`.

LumaBank — Data Dictionary

Auto-generated from `data/curated/` on 2026-09-11 · SEED=20260910. Do not edit by hand — run `python etl/gen_data_dictionary.py`.

17 curated tables.

`dim_acquisition_channel` • 6 rows

paid_social / organic / app_store_search / referral / influencer / unknown.

column	dtype	non-null	distinct	example
<code>channel_key</code>	object	100.0%	6	1
<code>channel_name</code>	object	100.0%	6	paid_social
<code>channel_group</code>	object	100.0%	4	paid
<code>is_paid</code>	object	100.0%	2	True

`dim_app_release` • 144 rows

Mobile + backend releases. `is_cache_reset_release` flags the 2026-07-16 release; a 2026-07-22 hotfix row follows; -1 = unknown.

column	dtype	non-null	distinct	example
<code>app_release_key</code>	object	100.0%	144	-1
<code>version</code>	object	100.0%	141	unknown
<code>platform</code>	object	100.0%	3	unknown
<code>released_at</code>	object	99.3%	122	2025-01-01 00:00:00
<code>released_date_key</code>	object	99.3%	122	20250101.0
<code>rollout_pct</code>	object	99.3%	4	90.0
<code>release_type</code>	object	100.0%	3	unknown
<code>notes</code>	object	100.0%	9	
<code>is_cache_reset_release</code>	object	100.0%	2	False

`dim_date` • 1,309 rows

Calendar spine, 2024-06-01 to 2027-12-31. Carries `experiment_phase` (pre / original / washout / rerun / post), `experiment_day`, `is_post_deploy`, and BR holiday / season attributes.

column	dtype	non-null	distinct	example
<code>date</code>	object	100.0%	1,309	2024-06-01 00:00:00
<code>date_key</code>	object	100.0%	1,309	20240601
<code>year</code>	object	100.0%	4	2024
<code>quarter</code>	object	100.0%	4	2
<code>month</code>	object	100.0%	12	6
<code>month_key</code>	object	100.0%	43	202406
<code>month_name</code>	object	100.0%	43	Jun 2024
<code>month_sort</code>	object	100.0%	43	24294
<code>iso_week</code>	object	100.0%	53	22

column	dtype	non-null	distinct	example
day_of_week	object	100.0%	7	5
is_month_end	object	100.0%	2	False
is_weekend	object	100.0%	2	True
is_school_break	object	100.0%	2	False
is_holiday_br	object	100.0%	2	False
season_tag	object	100.0%	4	Winter
experiment_phase	object	100.0%	6	out_of_window
is_post_deploy	object	100.0%	2	False
experiment_day	object	3.6%	28	1

dim_experiment_arm • 3 rows

Slicer helper: control / treatment / ambiguous.

column	dtype	non-null	distinct	example
arm_key	object	100.0%	3	control

dim_kyc_outcome • 9 rows

Distinct (outcome, reason_code, reason_group); is_rejection; -1 = none.

column	dtype	non-null	distinct	example
kyc_outcome_key	object	100.0%	9	-1
outcome	object	100.0%	4	none
reason_code	object	100.0%	7	none
reason_group	object	100.0%	5	n/a
is_rejection	object	100.0%	2	False

dim_onboarding_step • 6 rows

The six funnel steps in order.

column	dtype	non-null	distinct	example
step_key	object	100.0%	6	1
step_name	object	100.0%	6	signup_started
step_order	object	100.0%	6	1
funnel_stage	object	100.0%	5	signup

dim_region • 28 rows

27 UFs + macro region + timezone + utc_offset + is_amazon_tz; -1 = unknown. The axis the post-deploy fallback bias correlates with.

column	dtype	non-null	distinct	example
region_key	object	100.0%	28	-1
uf	object	100.0%	28	??
uf_name	object	100.0%	28	unknown
macro_region	object	100.0%	6	unknown

column	dtype	non-null	distinct	example
timezone	object	100.0%	9	America/Sao_Paulo
utc_offset_hours	object	100.0%	3	-3
is_amazon_tz	object	100.0%	2	False
pop_weight	object	100.0%	26	0.0

dim_user • 135,001 rows

One row per signup. Acquisition channel, region, device, app version at signup, and the derived experiment fields: effective arm, cohort (pre_deploy / post_deploy / rerun), contaminated_flag, in_analysis_flag.

column	dtype	non-null	distinct	example
user_key	object	100.0%	135,001	1
user_id	object	100.0%	135,001	LB1058661
signup_date_key	object	100.0%	589	20251219
signup_cohort_week	object	100.0%	93	2025-12-15
channel_key	object	100.0%	7	3
acquisition_channel	object	100.0%	6	app_store_search
region_key	object	100.0%	28	1
device_os	object	100.0%	3	Android
app_version_at_signup	object	100.0%	122	4.74.0
age_band	object	100.0%	1	unknown
birth_year_valid	object	100.0%	1	False
is_test_account	object	100.0%	1	False
experiment_arm	object	100.0%	4	not_in_experiment
experiment_cohort	object	100.0%	5	not_in_experiment
is_experiment_subject	object	100.0%	2	False
contaminated_flag	object	100.0%	2	False
assignment_switch_flag	object	100.0%	2	False
in_analysis_flag	object	100.0%	2	False

fact_activation • 135,000 rows

ONE ROW PER SIGNUP, platform-wide (is_experiment_subject flags the randomized ones). Effective arm, cohort, Day-1/7/30 activation, TTFT, and the guardrail flags (kyc_rejected, fraud_flag_30d, onboarding_tickets_n) pre-computed. in_analysis_flag = the clean re-run population.

column	dtype	non-null	distinct	example
user_key	object	100.0%	135,000	1
user_id	object	100.0%	135,000	LB1058661
arm_key	object	100.0%	3	not_in_experiment
assigned_date_key	object	100.0%	588	20251219
assigned_ts	object	100.0%	112,922	2025-12-19 18:27:00
experiment_cohort	object	100.0%	4	not_in_experiment
channel_key	object	100.0%	6	3

column	dtype	non-null	distinct	example
region_key	object	100.0%	27	1
app_version_at_signup	object	100.0%	121	4.74.0
account_opened_flag	object	100.0%	2	True
account_opened_ts	object	71.6%	95,975	2025-12-20 01:17:07.778368
first_txn_ts	object	50.2%	67,693	2025-12-22 00:27:27.643000
hours_to_first_txn	object	50.2%	64,331	54.008
day7_activated	object	100.0%	2	1
day1_activated	object	100.0%	2	0
day30_activated	object	100.0%	2	1
kyc_submitted_flag	object	100.0%	2	True
kyc_rejected_flag	object	100.0%	2	False
fraud_flag_30d	object	100.0%	2	False
onboarding_tickets_n	object	100.0%	4	0
is_experiment_subject	object	100.0%	2	False
contaminated_flag	object	100.0%	2	False
assignment_switch_flag	object	100.0%	2	False
in_analysis_flag	object	100.0%	2	False

fact_fraud_event • 944 rows

One row per fraud signal (flagged_ts in BRT), days_since_activation.

column	dtype	non-null	distinct	example
user_key	object	100.0%	944	131335
user_id	object	100.0%	944	LB1000115
flagged_ts	object	100.0%	944	2025-01-15 05:40:02
flagged_date_key	object	100.0%	495	20250115
signal_type	object	100.0%	4	synthetic_id
score	object	100.0%	397	0.899
is_confirmed	object	100.0%	2	True
days_since_activation	object	99.2%	30	12.0

fact_kyc_decision • 107,028 rows

One row per user's final KYC decision, joined to the submission (doc_type, submitted_after_account = the treatment-flow signature).

column	dtype	non-null	distinct	example
user_key	object	100.0%	105,447	76053
user_id	object	100.0%	107,028	LB1000002
kyc_outcome_key	object	100.0%	8	1
submitted_ts	object	98.5%	93,191	2025-01-01 11:30:00
decided_ts	object	100.0%	84,882	2025-01-01 00:00:00+00:00
decision_latency_hours	object	98.5%	48,179	-11.5

column	dtype	non-null	distinct	example
is_rejection	object	100.0%	2	False
is_manual_review	object	100.0%	2	False
doc_type	object	98.5%	2	rg
submitted_after_account	object	98.5%	2	False

fact_onboarding_step • 626,520 rows

One row per user x step (first occurrence) with seconds_since_signup.

column	dtype	non-null	distinct	example
user_key	object	100.0%	135,000	71277
user_id	object	100.0%	135,000	LB1049278
step_key	object	100.0%	6	3
step_name	object	100.0%	6	kyc_upload
occurred_ts	object	100.0%	491,360	2025-11-12 21:13:00
occurred_date_key	object	100.0%	665	20251112
seconds_since_signup	object	100.0%	247,669	900.0
is_reached	object	100.0%	1	True

fact_srm_daily • 44 rows

ONE ROW PER EXPERIMENT x CALENDAR DAY — the integrity-monitor surface. Daily arm counts, cumulative + trailing-7d chi-square SRM p-values, snapshot-vs-log mismatch count, alert_state. The 2026-07-16 break is here.

column	dtype	non-null	distinct	example
experiment_phase	object	100.0%	2	original
assign_date	object	100.0%	44	2026-07-06
assign_date_key	object	100.0%	44	20260706
n_control	object	100.0%	35	138
n_treatment	object	100.0%	35	126
n_fallback	object	100.0%	7	0
n_primary	object	100.0%	41	264
srm_chisq_cumulative	object	100.0%	41	0.545
srm_p_cumulative	object	100.0%	41	0.46
srm_chisq_trailing7	object	100.0%	41	0.545
srm_p_trailing7	object	100.0%	42	0.46
snapshot_log_mismatch_n	object	100.0%	11	3
multi_assignment_n	object	100.0%	11	12
alert_state	object	100.0%	2	ok

fact_support_ticket • 20,324 rows

One row per ticket; is_onboarding drives the support-cost guardrail.

column	dtype	non-null	distinct	example
user_key	object	100.0%	17,722	54428

column	dtype	non-null	distinct	example
user_id	object	100.0%	17,722	LB1000003
opened_ts	object	100.0%	20,324	2025-01-01 13:50:32.778499600
opened_date_key	object	100.0%	656	20250101
category	object	100.0%	5	onboarding
channel	object	100.0%	3	email
cidade	object	100.0%	4,333	Cirino De Goiás
is_onboarding	object	100.0%	2	True

fact_targets_month • 105 rows

Plan / target values per month per KPI (doc 01 section 15).

column	dtype	non-null	distinct	example
month_key	object	100.0%	21	202501
kpi	object	100.0%	5	day7_activation_rate
target_value	object	100.0%	45	0.44

fact_transaction_day • 2,002,414 rows

User x local day: transaction count, BRL, per-type counts, is_first_txn_day.

column	dtype	non-null	distinct	example
user_key	object	100.0%	67,705	89499
user_id	object	100.0%	67,705	LB1004300
date_key	object	100.0%	638	20250203
txn_count	object	100.0%	8	1
txn_amount_brl	object	100.0%	110,781	65.67
pix_count	object	100.0%	7	1
card_count	object	100.0%	6	0
boleto_count	object	100.0%	5	0
transfer_count	object	100.0%	6	0
is_first_txn_day	object	100.0%	2	True

fact_variant_assignment • 13,709 rows

One row per assignment EVENT (contaminated users have >=2). assignment_source (primary / fallback), cache_reset_flag, is_first_assignment, is_effective_assignment (the auditable rule).

column	dtype	non-null	distinct	example
user_key	object	100.0%	13,577	46613
user_id	object	100.0%	13,577	LB1108431
arm_key	object	100.0%	2	control
assigned_ts_brt	object	100.0%	10,657	2026-07-06 03:09:00
assigned_date_key	object	100.0%	41	20260706
assignment_seq	object	100.0%	2	1
assignment_source	object	100.0%	2	primary

column	dtype	non-null	distinct	example
cache_reset_flag	object	100.0%	2	False
app_release_key	object	100.0%	14	113
is_first_assignment	object	100.0%	2	True
is_effective_assignment	object	100.0%	2	True

LumaBank — ETL Pipeline

Document status: Phase 4 deliverable. **Depends on:** `02_data_architecture.md`. **Describes:** the code under `etl/` — what each script does, in what order, and the guarantees it gives.

PT. Este documento descreve o pipeline `etl/` já implementado: quatro scripts (`01_generate_raw` → `02_clean_stage` → `03_build_curated` → `dq_checks`), o que cada um garante, e como rodar. Um único `SEED` em `config.py` torna tudo reproduzível byte-a-byte — **incluindo a quebra de 16/07/2026**, que é determinística, não ruído.

1. Run

```
python -m pip install -r etl/requirements.txt
python etl/run_pipeline.py           # 01 -> 02 -> 03 -> dq_checks (~9 min)
python etl/run_pipeline.py --from 02 # skip regeneration
python etl/run_pipeline.py --from 03 # rebuild curated + dq only (~90s)
python etl/gen_data_dictionary.py    # regenerate docs/03 from data/curated/
```

`run_pipeline.py` runs each step as a subprocess and times it. `dq_checks.py` exits non-zero on any failed **hard** check, so a broken build fails loudly rather than publishing a silently-wrong `data/quality/dq_report.md`.

Determinism. `config.SEED` seeds `utils.rng(name)`, which derives one independent `numpy` Generator per named stream (`"world"`, `"txn"`, `"releases"`, `"raw_emit"`, ...), and `utils.stable_hash_u01(ids, salt)` derives a reproducible uniform draw per user id + salt — used for the feature-flag arm hash and for every step of the production bug (cache-reset roll, re-hash, fallback roll, fallback geo-skew). Same seed → byte-identical raw exports → identical curated tables, break included.

Idempotency. Each script wipes and rebuilds its own layer. `utils._atomic_write` retries on Windows/OneDrive file locks (`WinError 5/32`) with backoff.

2. `01_generate_raw.py` — the world simulator

Builds the whole synthetic world **in memory** as flat NumPy arrays (one row per user), then serialises it to six source-shaped, messy exports.

2.1 Users, timing, geography

`build_world` draws 135,000 signups against `SIGNUP_MONTH_WEIGHTS` (21 months, launch ramp + Jan/Nov-Dec seasonality), assigns UF/macro-region/timezone from `vocab/br_uf_region_tz.csv` (population-weighted), OS, acquisition channel (with a CRM-go-live gap → `unknown` for months 1–2), and a latent **intent** ~ $N(0,1)$ shifted by channel — the single confound that drives KYC-submission, activation and (mildly) fraud propensity throughout.

2.2 The experiment and the break

- **Membership.** `in_exp1` = signup in `[EXP1_START, EXP1_ENROLL_PAUSED]`; `in_exp2` = signup in `[EXP2_START, EXP2_END]`. `pre_deploy` / `post_deploy` split exp1 at `DEPLOY_DATE`.
- **Base assignment.** `stable_hash_u01(user_id, "...-v1")` for the original run, a **fresh** salt `"...-v2"` for the re-run — no arm carries over between runs.
- **BUG01 (cache reset).** For pre-deploy users who "open the app" (an 85% roll) within `BUG.reset_open_window_days` of `DEPLOY_DATE`, a `reset_share` roll marks a `cache_reset` exposure event.
- **BUG02 (contamination).** Of the reset users, a `reswitch_share` roll gives them a **second** `variant_assignments` row, re-hashed with a different salt — landing in either arm independently of the first (so ~half actually switch).
- **BUG03 (biased fallback).** For post-deploy signups, a `fallback_share` roll (constant over the ~6 remaining enrollment days, since the incident pauses enrollment before it would naturally decay) routes them to `assignment_source='fallback'`, whose arm is `stable_hash_u01(user_id, "fallback-geo") < fallback_treat_frac_by_region[macro_region]` — deterministic on geography, not the RNG.
- All of this is emitted as **assignment events**, not resolved arms — the resolution (effective arm, contamination) happens in curated (\$4), from the event log alone, exactly as the real monitoring pipeline would see it.

2.3 Onboarding, KYC, activation — the identity that makes Day-7 exact

Two funnels are computed side by side:

1. **Counterfactual control-flow** (`cf_*`) — arm-independent: would this user submit KYC fast, get approved, and have a decision inside 7 days? This drives `day7_base`, the **common baseline for both arms**, so the treatment effect is never a byproduct of the flow difference.
2. **Actual flow** (`is_treat_flow`) — control users get the counterfactual funnel exactly; treatment users get **account-first** (immediate, ~94%), KYC **deferred** (submitted hours later, a `treatment_limited_never_verified_share` never submit at all), and a `kyc_reject_delta_pp` / `fraud_flag_delta_pp` bump on the guardrails. This actual funnel drives every emitted onboarding-step / KYC / account row.

`day7` starts as `day7_base`, then a **deterministic largest-remainder flip**: for each (cohort, channel) cell of treatment subjects, `round(ATE_channel × n_cell)` not-yet-activated, day7-eligible users are flipped to activated, preferring the highest-intent candidates first. This makes the realised pooled and per-channel lift track `config.EFFECT` up to which-users noise — real binomial noise still comes from which half of the population landed in which (randomly assigned) arm.

The identity guarantee. Any user who ends up `day7` or a later (`late_act`) activator is **forced to have an account** (`acc_created |= day7 | late_act`) with a timestamp inside their flow's normal timing. This is why `dq_checks` can assert `day7_activated == account_opened_flag AND first_txn_ts <= assigned_ts + 7d` **exactly** — the curated recompute is not an approximation of the generator's truth, it reproduces it.

TTFT is drawn from a log-normal (median 26h) capped at 167.5h for Day-7 activators, and from a uniform(169h, 720h) for day-8-to-30 late activators; transactions past `WINDOW_END` are dropped (near-future censoring) rather than fabricated.

2.4 Messy exports

Six source systems (`lumencore`, `flagfox`, `trilha`, `riskguard`, `beacon`, `orbita`), each with its own quirks — see `02_data_architecture.md` §4–5 for the full inventory. Highlights: monthly user snapshots with intra-month duplicate rows; overlapping-month transaction files (cross-file dupes); `amount` as "R\$ 1.234,56" on ~6% of rows; transaction timestamps as Unix epoch-milliseconds; KYC decisions in UTC with ~20% date-only; app releases as Excel-serial dates in `.xlsx`; Latin-1 city names; at-least-once event logs.

3. `02_clean_stage.py` — raw → `stg_*`

One typed table per raw entity. `parse_dt_flex` handles ISO-8601 (mixed with/without fractional seconds — `format="ISO8601"`, not the pandas default, which silently returns `NaT` on a mixed-precision column), `DD/MM/YYYY` / `MM/DD/YYYY`, Excel serials, and epoch milliseconds/seconds all in one call, dispatching on the string shape.

Fix	How	Counter(s)
Date/time	<code>parse_dt_flex</code> ; RiskGuard <code>decided_at</code> additionally parsed as UTC with <code>format="ISO8601"</code> (mixed full-timestamp / date-only)	<code>dates_parsed</code> , <code>epochs_or_serials_converted</code> , <code>decisions_date_only</code>
Money	strip R\$ / thousands <code>.</code> ; <code>,</code> → <code>.</code>	<code>money_strings_parsed</code>
Encoding	<code>ftfy.fix_text</code> on Latin-1-damaged names / <code>cidade</code>	<code>mojibake_repaired</code>
De-dup	exact-duplicate rows on the natural key per entity (<code>txn_id</code> , <code>event_id</code> , <code>ticket_id</code> , ...); the monthly user snapshot collapses to one row per user (<code>keep="last"</code>)	<code>dupe_txn_dropped</code> , <code>dupe_events_dropped</code> , <code>snapshot_rows_collapsed</code>
Not de-duped	<code>variant_assignments</code> rows are kept as-is across <code>assignment_seq</code> — a second row per user is the contamination signal, not noise	<code>multi_assignment_users</code> (counted, not dropped)
Category	<code>step_name</code> , transaction <code>type</code> , ticket <code>category</code> , acquisition <code>channel</code> , release <code>platform</code> , KYC reason → controlled vocab in <code>etl/vocab/</code>	<code>*_normalised</code> , <code>*_unknowns</code>
Boolean	<code>cache_reset</code> , <code>confirmed</code> , <code>submitted_after_account</code> from "true"/"1" text	—

Staging performs **no** cross-source joins, no arm resolution, and no metric logic.

4. `03_build_curated.py` — `stg_*` → the star

- `dim_date` stamps `experiment_phase` (pre / original / washout / rerun / post), `experiment_day` (1..N within each run), and `is_post_deploy` (the original run, on/after the deploy).
- `dim_user` — one row per signup, joined to acquisition channel, region, and the **experiment derivation** below.
- **Effective arm + contamination (vectorised, not a per-row loop)**. From `fact_variant_assignment` 's event log: `first_arm` = the seq-1 arm; if a seq-2 row exists with a **different** arm **and** it lands before 50% of the user's Day-7 window has elapsed, that later arm is effective instead. `contaminated_flag` = an arm switch **or** a `cache_reset` exposure inside the active Day-7 window — the exact auditable rule from doc 02 §7.3, reproduced here as `pandas` boolean algebra over one `per`-user frame (no Python-level loop over 135k rows).
- `experiment_cohort` from the user's local signup date against the run boundaries; `in_analysis_flag` = `rerun` **and** not contaminated **and** never touched the fallback (`assignment_source` == 'primary' on every one of the user's rows) — the exact population the primary read-out filters on.
- `fact_activation` — one row per randomized user; `day7_activated` / `day1_activated` / `day30_activated` **recomputed from `fact_transaction_day` and the account-open timestamp**, not carried over from the generator — this is what makes the `dq_checks` identity assertion meaningful.
- `fact_srm_daily` — for each of the two runs, walks every calendar day and computes, from `fact_variant_assignment` alone: daily arm counts, a **cumulative** and a **trailing-7-day** Pearson χ^2 goodness-of-fit vs 50/50 (`scipy.stats.chi2.sf`), `snapshot_log_mismatch_n` / `multi_assignment_n` (cumulative to date), and `alert_state` per the doc 01 §9 runbook thresholds (`WARN` `p` < 0.01, `ALERT` `p` < 0.001, on either test).
- **Sentinel keys**. `user_key` = -1 for any event whose `user_id` is absent from `dim_user` (queue-lag orphans). Facts are never dropped for a missing dimension.
- **User-local calendar day**. UTC timestamps (Flagfox, RiskGuard) are converted to `America/Sao_Paulo` before bucketing (`to_brt_naive`); everything else is emitted already-local by the generator. *(Simplification flagged here: bucketing uses BRT for every user rather than each region's own timezone — the Amazon-timezone nuance from doc 02 §7.3 is preserved as a user attribute (`dim_region.timezone`) but not used for day-bucketing. Immaterial to the SRM signal, which is driven by the fallback's geographic skew, not by which clock a day boundary uses.)*

5. `dq_checks.py`

Hard assertions (build fails on violation): referential integrity on every fact; `fact_variant_assignment` unique on (`user_id`, `assignment_seq`); exactly one `is_first_assignment` and one `is_effective_assignment` per user; the Day-7 identity recompute; core-funnel step-reach monotonicity (`signup` → ... → `first_transaction` ; `limits_lifted` is excluded — it is a parallel KYC-completion milestone, not downstream of activation); **the SRM check** — the original run must hit `alert_state` == 'alert' on or after 2026-07-17 and raise `WARN`/`ALERT` on ≥ 2 days, while the re-run's trailing-7-day SRM p-value never drops below 0.01; every `in_analysis_flag` user is non-contaminated, `rerun` cohort, and never touched the fallback; no transaction-day rows past `WINDOW_END` ; every re-run user's Day-7 window closes inside the data.

Soft checks (reported, not fatal): whether the *cumulative* SRM p-value also crosses 0.001 (it doesn't, in this build — the trailing-7-day test is the one that fires, which is the point of running both); benchmark bands on the re-run control arm (Day-7 activation, KYC rejection, flagged-fraud, tickets/1k).

Current run: 0 hard failures, 1 warning (see `data/quality/dq_report.md`).

6. Performance notes

Full pipeline ≈ 9 min on the dev box (`01` ≈ 7 min, `02` ≈ 40 s, `03` ≈ 80 s, `dq` a few seconds). Two traps worth naming for anyone extending this: **(1)** `pandas.to_datetime` on a column that mixes timestamps with and without fractional seconds silently returns `NaT` for the non-matching rows unless `format="ISO8601"` is passed — this broke `signup_ts` for every user on the first pass and is why every ISO parse call here pins the format explicitly; **(2)** a per-group `.agg(..., lambda s: (s == x).sum())` on a multi-million-row `groupby` is dramatically slower than pre-computing the boolean columns and summing them — `fact_transaction_day` 's per-type counts went from several minutes to a few seconds this way.

End of Phase 4 deliverable (ETL).

LumaBank — Data Model

Document status: Phase 4 deliverable. **Depends on:** 02_data_architecture.md , 03_data_dictionary.md (auto-generated). Describes the curated star as it will be imported into the Power BI semantic model (Phase 6).

PT. O star curado: matriz de barramento (fatos × dimensões), grão e medidas de cada fato, atributos de cada dimensão, e as relações físicas para o modelo semântico.

1. Bus matrix (facts × dimensions)

Fact ↓ / Dim →	date	user	acquisition_channel	region	app_release	kyc_outcome	onboarding_step	experiment_arm
fact_variant_assignment	• (assigned)	•			•			•
fact_onboarding_step	• (occurred)	•					•	
fact_kyc_decision	(via decided_ts)	•				•		
fact_activation	• (assigned)	•	•	•				•
fact_transaction_day	•	•						
fact_fraud_event	• (flagged)	•						
fact_support_ticket	• (opened)	•						
fact_srm_daily	• (assign_date)							(arm counts as measures)
fact_targets_month	• (month_key, month grain)							

• = direct relationship; (via ...) = reachable through another table, not a physical relationship. Every fact carries *_date_key (int YYYYMMDD) → dim_date[date_key] except fact_targets_month, which joins on month_key (int YYYYMM) at month grain, and fact_kyc_decision, which carries timestamps but no integer date key (decisions are analysed by latency and by joining to fact_activation's date, not on their own calendar axis).

2. Fact tables — grain & measures

Fact	Grain	Additive measures	Non-additive / semi-additive
fact_variant_assignment	one assignment event (≥2 for contaminated users)	count	arm_key, assignment_source, cache_reset_flag, is_first_assignment, is_effective_assignment
fact_onboarding_step	user × step (first occurrence)	count	step_key, seconds_since_signup, is_reached
fact_kyc_decision	one final KYC decision per user	count	kyc_outcome_key, decision_latency_hours, is_rejection, is_manual_review, doc_type, submitted_after_account
fact_activation	one row per signup — platform-wide (is_experiment_subject flags the randomized ones)	onboarding_tickets_n	day7_activated (primary), day1_activated, day30_activated, hours_to_first_txn, account_opened_flag, kyc_submitted_flag, kyc_rejected_flag, fraud_flag_30d, arm_key, experiment_cohort,

Fact	Grain	Additive measures	Non-additive / semi-additive
			is_experiment_subject , contaminated_flag , assignment_switch_flag , in_analysis_flag
fact_transaction_day	user × local day	txn_count , txn_amount_br1 , pix_count , card_count , boleto_count , transfer_count	is_first_txn_day
fact_fraud_event	one fraud signal	count	signal_type , score , is_confirmed , days_since_activation
fact_support_ticket	one ticket	count	category , channel , is_onboarding
fact_srm_daily	experiment × calendar day (44 rows)	n_control , n_treatment , n_fallback , n_primary	srm_p_cumulative , srm_p_trailing7 , srm_chisq_cumulative , srm_chisq_trailing7 , snapshot_log_mismatch_n , multi_assignment_n , alert_state
fact_targets_month	month × kpi	target_value	kpi

fact_activation is the primary read-out surface: Day-7 Activation, every guardrail, and in_analysis_flag are pre-computed so the report needs almost no DAX and Phase 10 goes straight to the statistics. fact_srm_daily is the integrity-monitor surface: the whole "Experiment Integrity Monitor" page reads from this one table plus dim_date .

3. Dimensions — key attributes

Dimension	Key	Attributes used in the report
dim_date	date_key (int YYYYMMDD)	year , quarter , month , month_name , iso_week , day_of_week , is_weekend , is_holiday_br , is_school_break , season_tag , experiment_phase (pre / original / washout / rerun / post), experiment_day , is_post_deploy
dim_user	user_key	signup_date_key , signup_cohort_week , channel_key , acquisition_channel , region_key , device_os , app_version_at_signup , experiment_arm , experiment_cohort , is_experiment_subject , contaminated_flag , assignment_switch_flag , in_analysis_flag
dim_acquisition_channel	channel_key	channel_name (paid_social / organic / app_store_search / referral / influencer / unknown), channel_group (paid / owned / earned), is_paid
dim_region	region_key	uf (27), uf_name , macro_region , timezone , utc_offset_hours , is_amazon_tz — the axis the post-deploy fallback bias correlates with; -1 = unknown
dim_app_release	app_release_key	version , platform , released_at , rollout_pct , release_type , is_cache_reset_release (true only for the 2026-07-16 row); -1 = unknown
dim_kyc_outcome	kyc_outcome_key	outcome (approved / manual_review / rejected), reason_code , reason_group , is_rejection ; -1 = none
dim_onboarding_step	step_key	step_name (signup_started → personal_data → kyc_upload → account_created → first_transaction → limits_lifted), step_order , funnel_stage
dim_experiment_arm	arm_key	control / treatment / ambiguous — disconnected slicer helper

Sentinel keys. user_key = -1 for any event whose user_id is absent from dim_user (a queue-lag orphan in KYC or fraud decisions). Facts are never dropped for a missing dimension row.

4. Relationships

From (fact)	Column	To	Column	Card.	Cross-filter	Active
fact_variant_assignment	assigned_date_key	dim_date	date_key	* : 1	single	yes

From (fact)	Column	To	Column	Card.	Cross-filter	Active
fact_variant_assignment	user_key	dim_user	user_key	*:1	single	yes
fact_variant_assignment	app_release_key	dim_app_release	app_release_key	*:1	single	yes
fact_onboarding_step	occurred_date_key	dim_date	date_key	*:1	single	yes
fact_onboarding_step	user_key	dim_user	user_key	*:1	single	yes
fact_onboarding_step	step_key	dim_onboarding_step	step_key	*:1	single	yes
fact_kyc_decision	user_key	dim_user	user_key	*:1	single	yes
fact_kyc_decision	kyc_outcome_key	dim_kyc_outcome	kyc_outcome_key	*:1	single	yes
fact_activation	assigned_date_key	dim_date	date_key	*:1	single	yes
fact_activation	user_key	dim_user	user_key	1:1	single	yes
fact_activation	channel_key	dim_acquisition_channel	channel_key	*:1	single	yes
fact_activation	region_key	dim_region	region_key	*:1	single	yes
fact_transaction_day	date_key	dim_date	date_key	*:1	single	yes
fact_transaction_day	user_key	dim_user	user_key	*:1	single	yes
fact_fraud_event	flagged_date_key	dim_date	date_key	*:1	single	yes
fact_fraud_event	user_key	dim_user	user_key	*:1	single	yes
fact_support_ticket	opened_date_key	dim_date	date_key	*:1	single	yes
fact_support_ticket	user_key	dim_user	user_key	*:1	single	yes
fact_srm_daily	assign_date_key	dim_date	date_key	*:1	single	yes
fact_targets_month	month_key	dim_date	month_key (not the PK)	*:*	—	no physical relationship — joined in DAX via TREATAS against dim_date[month_key]

fact_activation is 1:1 with dim_user restricted to is_experiment_subject = true — every randomized user has exactly one row (§ dq_checks). dim_experiment_arm is a disconnected table: the report's arm slicer writes to a measure that filters fact_activation[arm_key] / fact_srm_daily[n_control|n_treatment] directly, not through a relationship — this is what lets one page show *both* arms side by side without a row-splitting filter.

```

erDiagram
    dim_date ||--o{ fact_variant_assignment : "assigned"
    dim_user ||--o{ fact_variant_assignment : ""
    dim_app_release ||--o{ fact_variant_assignment : ""
    dim_date ||--o{ fact_onboarding_step : "occurred"
    dim_user ||--o{ fact_onboarding_step : ""
    dim_onboarding_step ||--o{ fact_onboarding_step : ""
    dim_user ||--o{ fact_kyc_decision : ""
    dim_kyc_outcome ||--o{ fact_kyc_decision : ""
    dim_date ||--o{ fact_activation : "assigned"
    dim_user ||--|| fact_activation : "1:1 (subjects)"
    dim_acquisition_channel ||--o{ fact_activation : ""
    dim_region ||--o{ fact_activation : ""
    dim_date ||--o{ fact_transaction_day : ""
    dim_user ||--o{ fact_transaction_day : ""
    dim_date ||--o{ fact_fraud_event : "flagged"
    dim_user ||--o{ fact_fraud_event : ""
    dim_date ||--o{ fact_support_ticket : "opened"
    dim_user ||--o{ fact_support_ticket : ""
    dim_date ||--o{ fact_srm_daily : "assign_date"

```

5. Measure groups (Phase 7 preview)

Activation & Funnel (Day-7/1/30 Activation, TTFT median/p90, funnel step-reach and conversion by flow) · **Acquisition** (signups and Day-7 Activation by channel) · **Risk & Guardrails** (KYC rejection rate, manual-review rate, flagged-fraud rate, onboarding tickets/1k — all with non-inferiority framing by arm) · **Experiment Integrity** (daily allocation by arm, cumulative + trailing-7d SRM p-value, alert_state, snapshot/log mismatch, contaminated-vs-clean counts) · **Experiment Readout** (Day-7 Activation by arm + lift + CI + p, guardrail deltas + NI test, heterogeneity by channel + interaction test, power/MDE context) · **Platform Health** (active users, transaction volume, trend) · **Time Intelligence** (WoW, trailing periods) · **Targets vs Plan**.

End of Phase 4 deliverable (Data Model).

LumaBank — DAX Measure Library

Auto-generated from `etl/pbi_measures.py` on 2026-09-10 by `etl/gen_dax_doc.py`. Do not edit by hand.

46 measures in the `_Measures` table, injected into the semantic model by `powerbi/gen_semantic_model.py`.

Conventions

- `fact_activation` is **one row per signup, platform-wide** — most measures here carry no arm filter (platform-health questions); the Experiment Readout group filters `in_analysis_flag = TRUE()` (the clean re-run population only).
- "Latest day" = `MAX(dim_date[date])` in the current filter context. No marked date table, no time-intelligence functions — trailing windows use explicit `dim_date[date]` arithmetic.
- SRM measures read `fact_srm_daily` (one row per experiment x calendar day) and are never sliced by arm.
- KYC rejection rate, fraud flag rate, and onboarding tickets are lower-is-better** — guardrail delta measures are framed so positive = treatment worse, matching the non-inferiority bounds in doc 01 section 8.3 (+1.5 pp KYC, +0.5 pp fraud).

01 Platform Health

Signups

format `#,0`

One row per signup (platform-wide) — trends by `dim_date` via `assigned_date_key`.

```
COUNTROWS ( fact_activation )
```

Active Users

format `#,0`

Users with ≥ 1 transaction in the trailing 30 days ending at the latest date in context.

```
VAR maxd = MAX ( dim_date[date] )
VAR lo    = maxd - 29
RETURN
CALCULATE (
    DISTINCTCOUNT ( fact_transaction_day[user_key] ),
    dim_date[date] >= lo, dim_date[date] <= maxd,
    REMOVEFILTERS ( fact_activation )
)
```

Active Rate %

format `0.0%`

```
DIVIDE ( [Active Users], [Signups] )
```

Account-Creation Rate %

format `0.0%`

Signups that ever reach an account, any state.

```
DIVIDE (
    CALCULATE ( COUNTROWS ( fact_activation ), fact_activation[account_opened_flag] = TRUE ( ) ),
    [Signups]
)
```

02 Onboarding Funnel

Funnel Reach

format `#,0`

Users reaching the step selected by the Onboarding Step slicer.

```
DISTINCTCOUNT ( fact_onboarding_step[user_key] )
```

Funnel Step Conversion %

format `0.0%`

Funnel Reach as a share of all signups — plot by dim_onboarding_step[step_order].

```
DIVIDE ( [Funnel Reach], [Signups] )
```

KYC-Submit Rate %

format `0.0%`

```
DIVIDE (
    CALCULATE ( COUNTROWS ( fact_activation ), fact_activation[kyc_submitted_flag] = TRUE () ),
    [Signups]
)
```

Median Minutes Signup to Account

format `#,0`

```
MEDIANX (
    FILTER ( fact_activation, fact_activation[account_opened_flag] = TRUE () ),
    DATEDIFF ( fact_activation[signup_ts], fact_activation[account_opened_ts], MINUTE )
)
```

Limited Never-Verified % (Re-run Treatment)

format `0.0%`

Treatment subjects (clean re-run) with an account who never submitted KYC.

```
VAR base =
    CALCULATE ( COUNTROWS ( fact_activation ),
        fact_activation[in_analysis_flag] = TRUE (), fact_activation[arm_key] = "treatment" )
VAR nv =
    CALCULATE ( COUNTROWS ( fact_activation ),
        fact_activation[in_analysis_flag] = TRUE (), fact_activation[arm_key] = "treatment",
        fact_activation[account_opened_flag] = TRUE (), fact_activation[kyc_submitted_flag] = FALSE () )
RETURN DIVIDE ( nv, base )
```

03 Activation

Day-7 Activation Rate %

format `0.0%`

PRIMARY METRIC. Account created and first transaction within 7 days of signup.

```
DIVIDE ( SUM ( fact_activation[day7_activated] ), COUNTROWS ( fact_activation ) )
```

Day-1 Activation Rate %

format `0.0%`

```
DIVIDE ( SUM ( fact_activation[day1_activated] ), COUNTROWS ( fact_activation ) )
```

Day-30 Activation Rate %

format 0.0%

```
DIVIDE ( SUM ( fact_activation[day30_activated] ), COUNTROWS ( fact_activation ) )
```

Median Hours to First Transaction

format #,0

```
MEDIANX (
  FILTER ( fact_activation, NOT ISBLANK ( fact_activation[first_txn_ts] ) ),
  fact_activation[hours_to_first_txn]
)
```

P90 Hours to First Transaction

format #,0

```
PERCENTILEX.INC (
  FILTER ( fact_activation, NOT ISBLANK ( fact_activation[first_txn_ts] ) ),
  fact_activation[hours_to_first_txn], 0.9
)
```

04 Risk & Guardrails

KYC Rejection Rate %

format 0.0%

Rejected / KYC submissions. Non-negotiable guardrail.

```
DIVIDE (
  CALCULATE ( COUNTROWS ( fact_activation ), fact_activation[kyc_rejected_flag] = TRUE () ),
  CALCULATE ( COUNTROWS ( fact_activation ), fact_activation[kyc_submitted_flag] = TRUE () )
)
```

KYC Manual-Review Rate %

format 0.0%

```
DIVIDE (
  CALCULATE ( COUNTROWS ( fact_kyc_decision ), fact_kyc_decision[is_manual_review] = TRUE () ),
  COUNTROWS ( fact_kyc_decision )
)
```

Flagged-Fraud Rate % (30d)

format 0.00%

Confirmed fraud signals within 30 days / Day-7-activated accounts. Non-negotiable guardrail.

```
DIVIDE (
  CALCULATE ( COUNTROWS ( fact_activation ), fact_activation[fraud_flag_30d] = TRUE () ),
  CALCULATE ( COUNTROWS ( fact_activation ), fact_activation[day7_activated] = 1 )
)
```

05 Support & Cost

Onboarding Tickets

format `#,0`

```
SUM ( fact_activation[onboarding_tickets_n] )
```

Onboarding Tickets per 1k Signups

format `#,0.0`

```
DIVIDE ( [Onboarding Tickets], [Signups] ) * 1000
```

Onboarding Contact Rate %

format `0.0%`

```
DIVIDE (
  CALCULATE ( COUNTROWS ( fact_activation ), fact_activation[onboarding_tickets_n] > 0 ),
  [Signups]
)
```

06 Experiment Integrity

SRM P-Value (Cumulative)

format `0.0000`

Pearson chi-square goodness-of-fit vs 50/50, cumulative to the day in context.

```
AVERAGE ( fact_srm_daily[srm_p_cumulative] )
```

SRM P-Value (Trailing 7d)

format `0.0000`

The test that actually catches a mid-flight break — see docs/01 section 9.

```
AVERAGE ( fact_srm_daily[srm_p_trailing7] )
```

Allocation Ratio (Treatment Share)

format `0.0%`

Expect 50%. The daily line that shows the break.

```
DIVIDE (
  SUM ( fact_srm_daily[n_treatment] ),
  SUM ( fact_srm_daily[n_treatment] ) + SUM ( fact_srm_daily[n_control] )
)
```

Fallback Share %

format `0.0%`

```
DIVIDE (
  SUM ( fact_srm_daily[n_fallback] ),
  SUM ( fact_srm_daily[n_fallback] ) + SUM ( fact_srm_daily[n_primary] )
)
```

SRM Worst State (Period)

Worst alert_state reached anywhere in the current filter — colour this with ALERT_COLORS.

```
VAR hasAlert = CALCULATE ( COUNTROWS ( fact_srm_daily ), fact_srm_daily[alert_state] = "alert" ) > 0
VAR hasWarn = CALCULATE ( COUNTROWS ( fact_srm_daily ), fact_srm_daily[alert_state] = "warn" ) > 0
RETURN IF ( hasAlert, "ALERT", IF ( hasWarn, "WARN", "OK" ) )
```

Contaminated Users

format #,0

```
CALCULATE ( COUNTROWS ( fact_activation ), fact_activation[contaminated_flag] = TRUE () )
```

Snapshot-Log Mismatch (Cumulative)

format #,0

```
MAX ( fact_srm_daily[snapshot_log_mismatch_n] )
```

Multi-Assignment Users (Cumulative)

format #,0

```
MAX ( fact_srm_daily[multi_assignment_n] )
```

07 Experiment Readout

Exp Subjects

format #,0

The clean re-run population only.

```
CALCULATE ( COUNTROWS ( fact_activation ), fact_activation[in_analysis_flag] = TRUE () )
```

Exp Activation Rate

format 0.0%

```
DIVIDE (
    CALCULATE ( SUM ( fact_activation[day7_activated] ), fact_activation[in_analysis_flag] = TRUE () ),
    [Exp Subjects]
)
```

Exp Control Rate

format 0.0%

```
CALCULATE ( [Exp Activation Rate], fact_activation[arm_key] = "control" )
```

Exp Treatment Rate

format 0.0%

```
CALCULATE ( [Exp Activation Rate], fact_activation[arm_key] = "treatment" )
```

Exp Lift pp

format +0.00;-0.00;0.00

Absolute treatment - control difference in Day-7 Activation, in percentage points.

```
( [Exp Treatment Rate] - [Exp Control Rate] ) * 100
```

Exp Lift CI Low pp

format +0.00;-0.00;0.00

```
VAR p1 = [Exp Control Rate]  VAR p2 = [Exp Treatment Rate]
VAR n1 = CALCULATE ( [Exp Subjects], fact_activation[arm_key] = "control" )
VAR n2 = CALCULATE ( [Exp Subjects], fact_activation[arm_key] = "treatment" )
VAR se = SQRT ( DIVIDE ( p1 * ( 1 - p1 ), n1 ) + DIVIDE ( p2 * ( 1 - p2 ), n2 ) )
RETURN ( ( p2 - p1 ) - 1.959964 * se ) * 100
```

Exp Lift CI High pp

format +0.00;-0.00;0.00

```
VAR p1 = [Exp Control Rate]  VAR p2 = [Exp Treatment Rate]
VAR n1 = CALCULATE ( [Exp Subjects], fact_activation[arm_key] = "control" )
VAR n2 = CALCULATE ( [Exp Subjects], fact_activation[arm_key] = "treatment" )
VAR se = SQRT ( DIVIDE ( p1 * ( 1 - p1 ), n1 ) + DIVIDE ( p2 * ( 1 - p2 ), n2 ) )
RETURN ( ( p2 - p1 ) + 1.959964 * se ) * 100
```

Exp Z Stat

format 0.00

Two-proportion pooled z-statistic.

```
VAR p1 = [Exp Control Rate]  VAR p2 = [Exp Treatment Rate]
VAR n1 = CALCULATE ( [Exp Subjects], fact_activation[arm_key] = "control" )
VAR n2 = CALCULATE ( [Exp Subjects], fact_activation[arm_key] = "treatment" )
VAR pp = DIVIDE ( p1 * n1 + p2 * n2, n1 + n2 )
VAR se = SQRT ( pp * ( 1 - pp ) * ( DIVIDE ( 1, n1 ) + DIVIDE ( 1, n2 ) ) )
RETURN DIVIDE ( p2 - p1, se )
```

Exp Significant 95%

```
IF ( ABS ( [Exp Z Stat] ) > 1.959964, "Significant", "Not significant" )
```

Exp KYC Rejection Delta pp

format +0.00;-0.00;0.00

Guardrail: non-inferiority margin is +1.5 pp (doc 01 section 8.3).

```
VAR r1 = CALCULATE ( [KYC Rejection Rate %], fact_activation[in_analysis_flag] = TRUE (), fact_activation[arm_key] = "control" )
VAR r2 = CALCULATE ( [KYC Rejection Rate %], fact_activation[in_analysis_flag] = TRUE (), fact_activation[arm_key] = "treatment" )
RETURN ( r2 - r1 ) * 100
```

Exp Fraud Delta pp

format +0.00;-0.00;0.00

Guardrail: non-inferiority margin is +0.5 pp (doc 01 section 8.3).

```
VAR r1 = CALCULATE ( [Flagged-Fraud Rate % (30d)], fact_activation[in_analysis_flag] = TRUE (), fact_activation[arm_key] = "control" )
VAR r2 = CALCULATE ( [Flagged-Fraud Rate % (30d)], fact_activation[in_analysis_flag] = TRUE (), fact_activation[arm_key] = "treatment" )
RETURN ( r2 - r1 ) * 100
```

Exp Tickets per 1k Delta

format +0.0;-0.0;0.0

Guardrail (secondary, directional).

```
VAR t1 = CALCULATE ( [Onboarding Tickets per 1k Signups], fact_activation[in_analysis_flag] = TRUE (), fact_activation[in_analysis_flag] = TRUE () )
VAR t2 = CALCULATE ( [Onboarding Tickets per 1k Signups], fact_activation[in_analysis_flag] = TRUE (), fact_activation[in_analysis_flag] = TRUE () )
RETURN t2 - t1
```

Exp Effect by Channel pp

format +0.00;-0.00;0.00

Same as Exp Lift pp; place on a bar by Acquisition Channel for the heterogeneity cut.

[Exp Lift pp]

08 Time & Targets

Day-7 Activation Target %

format 0.0%

```
CALCULATE ( SUM ( fact_targets_month[target_value] ), fact_targets_month[kpi] = "day7_activation_rate" )
```

Day-7 Activation vs Target pp

format +0.00;-0.00;0.00

```
( [Day-7 Activation Rate %] - [Day-7 Activation Target %] ) * 100
```

Active Users Target

format #,0

```
CALCULATE ( SUM ( fact_targets_month[target_value] ), fact_targets_month[kpi] = "active_users" )
```

KYC Rejection Target %

format 0.0%

A ceiling, not a target to beat — lower is not automatically better (doc 01 section 15).

```
CALCULATE ( SUM ( fact_targets_month[target_value] ), fact_targets_month[kpi] = "kyc_rejection_rate" )
```

09 Narrative

Exec Insight

A dynamic one-line summary for the Executive Overview page.

```
VAR d7      = [Day-7 Activation Rate %]
VAR kyc     = [KYC Rejection Rate %]
VAR fraud   = [Flagged-Fraud Rate % (30d)]
VAR srm     = [SRM Worst State (Period)]
RETURN
    "Day-7 Activation is " & FORMAT ( d7, "0.0%" ) &
```

```
". KYC rejection " & FORMAT ( kyc, "0.0%" ) & ", flagged fraud " & FORMAT ( fraud, "0.00%" ) &
". Experiment integrity: " & srm & "."
```

LumaBank — Visual Identity

Document status: Phase 5 deliverable. The single source of truth is `assets/brand.py` , imported by `assets/gen_logo.py` and `powerbi/gen_report_visuals.py` (Phase 8) so the logo, page backgrounds, theme, and every scripted visual override draw from the same values.

Tone

Trust-first fintech, not a marketing page — this is an incident read-out and an experiment integrity monitor as much as it is a dashboard. Deliberately **not** blue (AeroVanti, VoltEdge) and not violet or forest green (Voxa+, SwiftBite), and deliberately **not** money-green either.

Palette

Role	Hex	Use
Petrol (brand)	#0E4A52	brand, primary series, table accent
Petrol deep	#0A333A	header band, darkest ink
Coral (accent)	#E2624B	treatment arm · current period · the break · "look here"
Stone (comparison)	#93A29C	control arm · prior period · target
Page ground	#F3F1EC	warm off-white — not the sibling projects' cool blue-grey
Card	FFFFFF	
Gridline / hairline	#E9E5DC / #D8D2C4	quiet
Good / Bad / Neutral	#3C8073 / #C0453B / #8A8F8A	semantic; KYC rejection / fraud / tickets are lower-is-better

Categorical ramp: petrol, stone, coral, gold #C99A3E , plum #7A5C7E , slate #5B6B72 , dusty rose #C77C6E , deep blue #3B5B73 . Channel ramp (CHANNEL_COLORS): paid_social slate-blue, organic petrol, app_store_search plum, referral gold, influencer dusty rose, unknown grey. SRM alert_state ramp (ALERT_COLORS): ok teal-green, warn gold, alert bad-red — used on the Integrity Monitor page's daily strip.

Type

Segoe UI family. Semibold for titles / callouts, Regular for body / labels. Callout (KPI value) 30 px, title 12–15 px, label 9 px.

Signature element (every page)

Petrol header band (58 px) with the **LumaBank** wordmark ("Luma" ink, "Bank" coral), a coral 3 px hairline under it, a right-aligned "Activation Experiment — Fintech Analytics" label, then the page title + one-line question on the warm light body. Below: a slicer row (Cohort · Experiment arm · Acquisition channel), a white rounded KPI frame, then two content rows on the 3-30-300 gradient.

Assets emitted (assets/gen_logo.py → powerbi/assets/)

- `logo_mark_on_light.png` / `_on_dark.png` — rounded-square tile, a rising step-line ("activation") to a coral dot (the Day-7 activation moment) — not a vault, shield, or piggy-bank cliché.
- `wordmark_on_light.png` / `_on_dark.png` — "LumaBank" lockup.
- `bg_01_executive.png` (1280×860) · `bg_02_funnel.png` · `bg_03_integrity.png` · `bg_04_readout.png` (1280×720) — full-page backgrounds carrying the band + title.
- `_page_heights.json` .

Theme

`powerbi/theme/theme.json` — `dataColors` , semantic colours, 4 text classes. Lean, matching the AeroVanti-style "PBI lite" scope (Phase 0): no `visualStyles` block — per-visual formatting is set directly on each visual in Phase 8, not centralised in the theme.

End of Phase 5 deliverable.

LumaBank Activation — Report Structure

Document status: Phase 8 deliverable. The Power BI report is the **secondary** deliverable — the primary read-out is `data_analysis/.4` pages, code-generated. One command rebuilds it:

```
cd powerbi && bash build_report.sh
```

Pipeline: `gen_logo.py` → `gen_semantic_model.py` (+ `pbi_measures.py`) → `gen_dax_doc.py` → `gen_report_visuals.py` → `build_report.sh` (drives `pbir`). Output: `powerbi/LumaBankActivation.pbip` + `.Report` + `.SemanticModel`. `pbir validate --all` passes.

First open in Power BI Desktop: Transform data → set `pDataFolder` to `data/curated` → Close & Apply · View ▶ Themes ▶ Browse → `powerbi/theme/theme.json` .

Semantic model

18 imported curated tables + `_Measures` (46 measures in 9 folders, see `docs/06_dax_measures.md`), 22 relationships. `fact_activation` (one row per signup, platform-wide) drives Executive Overview and the funnel page; the Experiment Readout group filters to `in_analysis_flag = TRUE()` — the clean re-run population only, so the read-out never mixes the broken original run with the clean one.

The four pages

No on-canvas slicers — each page has a fixed analysis window baked into its measures (platform-wide vs. `in_analysis_flag` -filtered), matching the incident narrative rather than letting a slicer blend broken and clean data. Page backgrounds carry the header band + kicker + page title, generated by `gen_report_visuals.py` from `assets/brand.py` .

#	Page	Question	KPI row (4 cards)	Content
1	Executive Overview	Is the platform healthy, and did the redesign work?	Signups · Active Rate % · Day-7 Activation Rate % · Exp Lift pp	Day-7 Activation by channel (column) · Exec Insight narrative card · Day-7 Activation trend (line)
2	Onboarding & Activation Funnel	Where do new users fall out on the way to activation?	Account-Creation Rate % · Day-7 Activation Rate % · KYC-Submit Rate % · Median Minutes Signup→Account	Onboarding funnel (funnel chart) · KYC rejection by channel (column)
3	Experiment Integrity Monitor	Can the experiment be trusted right now — allocation, SRM, contamination?	Allocation Ratio (Treatment Share) · Contaminated Users · SRM P-Value (Trailing 7d) · SRM Worst State	Allocation vs. SRM p-value (line) · fallback-share trend (line) · snapshot-log mismatch trend (line) — <i>this page exists to tell the break story visually</i>
4	Experiment Readout	Once the experiment is clean, what is the effect?	Exp Control Rate · Exp Treatment Rate · Exp Lift pp · Exp Significant 95%	Effect by channel (column) · guardrails (multi-row card: KYC-rejection and flagged-fraud non-inferiority)

Known lite-scope gaps (not polished)

- No on-canvas slicers — filtering to the clean re-run population is done in DAX (`in_analysis_flag`), not by a user-facing control.
- No screenshot-verification loop (needs a live Desktop session).
- `Exec Insight` is a dynamic text measure (no Smart Narrative visual).

LumaBank Activation Experiment — Build Walkthrough

A running narrative of the build. Each entry: what was done, why, where to look. Newest at the bottom. Written as the phases complete, not reconstructed after the fact.

Phase 0 — Scope & scaffold (2026-09-10)

Locked scope in `docs/00_scope.md`. Key calls: analysis-primary (Phase 10 is the headline deliverable, following the AeroVanti model rather than VoltEdge/Voxa+/SwiftBite), Power BI "lite" (4 pages), fintech-specific guardrails (KYC rejection, flagged fraud) as non-negotiable, BRL only. The core hook — an onboarding A/B test that breaks mid-flight on a real production bug — is what the whole project exists to demonstrate, so the scope doc spends most of its length on the break mechanics (§4.2, §6) rather than on the product itself. Created the folder tree under `PBI Pessoal/LumaBank-Activation-Experiment/`.

Phase 1 — Business context & KPI framework (2026-09-10)

`docs/01_business_context_kpis.md`. Fictional Brazilian neobank; the two onboarding flows (control = document upload before account creation, treatment = account-first with KYC deferred to a limits gate) are specified before any data exists, exactly like AeroVanti's Flash Redemption spec. The centerpiece is §9: the **continuous SRM + assignment-integrity monitoring runbook** — five tests, WARN/ALERT thresholds, an ordered triage checklist. This is what makes the project different from AeroVanti's balance check: there, SRM is a single pre-flight test; here it's a standing process that has to *catch* something.

Phase 2 — Data architecture (2026-09-10)

`docs/02_data_architecture.md`. Medallion raw → staging → curated. Six simulated systems (Lumen Core Banking, Flagfox feature-flag service, Trilha onboarding, RiskGuard KYC/fraud, Beacon mobile-release log, Órbita growth CRM). The curated star's two load-bearing tables are `fact_activation` (the read-out surface) and `fact_srm_daily` (the integrity-monitor surface) — both fully spec'd here, before either exists.

Phase 3 — ETL pipeline (2026-09-10)

3.1 The true effect and the bug (kept only in `etl/config.py::EFFECT / BUG`)

Design intent: a real, channel-heterogeneous Day-7 Activation lift (paid_social/influencer biggest, referral/organic smallest) sized close enough to the re-run's MDE that the "deadline cost us precision" story actually bites. The break: a deploy-triggered cache reset (contamination) plus a region/timezone-biased fallback for post-deploy signups (composition bias) — both deterministic functions of a stable per-user hash, never the RNG directly, so the same seed always reproduces the same incident.

3.2 Pipeline shape

Same shape as the sibling projects: `config.py` → `utils.py` → `01_generate_raw.py` → `02_clean_stage.py` → `03_build_curated.py` → `dq_checks.py` → `run_pipeline.py`, plus `gen_data_dictionary.py`.

3.3 Fixes made during the build

- **pandas.to_datetime on mixed-precision ISO-8601 silently drops rows.** The generator emits onboarding-step timestamps as `.isoformat()`; `signup_started` events always land on a whole minute (no fractional seconds) while activation events carry microseconds. Parsing the column with the pandas default (`format=None`) infers ONE format from the first non-null value and returns `NaT` for every row that doesn't match — silently, no error. Every user's `signup_ts` came back null, which cascaded into a `KeyError: 'user_id'` three layers downstream (an empty `pd.DataFrame(rows)` had no columns to merge on) that took longer to trace than the actual bug. Fix: pin `format="ISO8601"` on every ISO parse call in `02_clean_stage.py` — it's the one format string that accepts mixed precision.
- **Lambda aggs on a multi-million-row groupby are dramatically slower than precomputed booleans.** `fact_transaction_day`'s per-type counts (`lambda s: (s == "pix").sum()` inside `.agg(...)`) took several minutes over ~2.7M transaction rows. Replaced with boolean columns computed once (`tx["pix_count"] = (tx["type"] == "pix").astype("int32")`) and summed inside a plain `.agg()` — seconds instead of minutes.
- **The treatment-effect double-count.** First cut computed `day7_eligible` for the baseline using each arm's OWN actual funnel (control's KYC-gated timing vs. treatment's account-first timing), then added the explicit per-channel ATE flips on top. Treatment's structurally higher account-creation rate (~94% vs. control's ~74%) was *already* an implicit ~10pp lift before a single flip was

applied — the realised pooled lift came out at **+12.8 pp** against a **+3.6 pp** target. Fixed by splitting the funnel into a **counterfactual control-flow** version (arm-independent, drives the common baseline `day7_base`) and the **actual flow** (drives the emitted onboarding/KYC/account data). The explicit ATE flips are now the *only* source of the treatment effect on Day-7.

- **...and then losing ~3 pp of it in the curated recompute.** After the fix above, the realised lift measured directly in the generator (`world["day7"]`) matched the +3.6pp target, but `fact_activation.day7_activated` — recomputed in curated from `account_opened_flag AND first_txn_ts <= assigned_ts + 7d`, independently of the generator's own flag, by design (`dq_checks` asserts the two agree) — showed only +1.1pp for the re-run. Cause: a flipped (explicitly-activated) treatment user was drawn from the counterfactual eligibility pool, but their *actual* account-creation roll (`t_acc_first`, ~94%) is an independent coin flip — about 9% of flipped users had no real account row, so the curated recompute correctly saw them as not activated. Fixed at the source: any user who ends up `day7` or a later (`late_act`) activator is now **forced** to have an account (`acc_created |= day7 | late_act`) with a timestamp inside their flow's normal window — the generator's intent and the curated recompute now agree exactly, by construction rather than by coincidence.
- **Calibrating the break to actually trip the SRM runbook's thresholds.** The first calibration pass (moderate fallback share, mild geographic skew) produced a detectable-but-weak signal: cumulative χ^2 p bottomed around 0.03, never crossing the 0.001 ALERT threshold, and the daily counts were too thin for the trailing-7-day test to fire cleanly either. Raised `fallback_share_0` (0.55 → 0.72) and sharpened the region-skew spread (`fallback_treat_frac_by_region` SE 0.66→0.78, N 0.31→0.35) so the bias is large enough, and geographically distinct enough, for the trailing-7-day χ^2 test to reach `p < 0.001` — the way the runbook actually catches things. Final calibration: ALERT fires on day 13 (2026-07-18, two days after the deploy); the cumulative test never crosses 0.001 (diluted by ten clean pre-deploy days), which is itself the right behaviour and is why the runbook (doc 01 59) runs both tests, not just one.
- **Near-future censoring.** A handful of late (day 8–30) activators from signups near the window's end would have their first transaction *after* `WINDOW_END` — the generator now drops those transaction rows rather than fabricating post-window history, so `day30_activated` for those users is correctly `0` (unobserved), not silently wrong.
- **dq_checks assertion tuned to the real detector.** Originally asserted the *cumulative* SRM p-value crosses 0.001 by 2026-07-17 — too strict once the fallback timing was recalibrated (ALERT fires on the trailing-7-day test one day later than that, which is the correct and intended detection path). Rewrote the hard check to require an `alert_state == "alert"` day on/after 2026-07-17 and downgraded "does the cumulative test also cross 0.001" to a soft, informational check.

Final run (SEED 20260910) — 0 hard DQ failures, 1 benign warning: 135,000 signups; re-run control Day-7 Activation 45.2%, treatment 49.8% (**lift +4.6 pp**, re-run MDE $\approx$ 3.0 pp); KYC rejection 8.1%, flagged-fraud 0.95%, onboarding tickets 94/1k; 191 contaminated + 681 biased-fallback users in the original run; SRM ALERT first fires 2026-07-18.

Phase 4 — Data documentation (2026-09-10)

`docs/03_data_dictionary.md` (auto-generated, 17 tables), `docs/04_etl_pipeline.md`, `docs/05_data_model.md` (bus matrix, 9 facts $\times$ 8 dims, mermaid ER diagram). No surprises here — the corrections that *do* touch these docs (the `fact_activation` grain change) were made later, in Phase 7, and backported into docs 02/05 at that time rather than re-litigated here.

Phase 5 — Visual identity (2026-09-10)

Proposed "deep ink-blue + coral" verbally before touching any file. Building `assets/brand.py`, checked the actual hex values of the four sibling palettes and found the proposed ink-blue sat close to VoltEdge's navy (`#0E2A4E` / `#123E6E`) and wasn't far from AeroVanti's blue (`#0B5FA5`) either — exactly the collision the brief said to avoid. Switched to **petrol/teal-ink** `#0E4A52` + **coral** `#E2624B` + **stone** `#93A29C`, warm off-white ground — verified distinct from all four siblings (2 $\times$ blue, 1 $\times$ violet, 1 $\times$ forest-green) before committing. `assets/gen_logo.py` draws the mark as a rising step-line to a coral dot (the Day-7 activation moment) rather than a vault/shield/piggy-bank cliché.

Phase 6 — Semantic model as code (TMDL) (2026-09-10)

`powerbi/gen_semantic_model.py` — same schema-introspection pattern as the sibling projects: read every curated Parquet, emit TMDL. 18 tables (17 curated + a `dim_month` helper scoped to the fact window), 22 relationships.

- **dim_app_release fan-out.** The bug-catalogue release (`version = "4.61.0"`) and its hotfix are emitted for *both* iOS and Android — two rows sharing one version string. The first relationship-building pass joined `fact_variant_assignment.app_version` to `dim_app_release.version` directly, which isn't unique, so the join fanned out and `fact_variant_assignment` inflated from 13,709 to 17,323 rows (breaking the "unique on (`user_id`, `assignment_seq`)" hard check). Fixed by deduplicating the release lookup on `version` before the merge — platform isn't needed for this join, only the release metadata is.
- **A real sentinel row, not just a schema-sample one.** The sibling projects' generators add a sentinel dimension row (e.g. `member_key = -1`) only to the 50-row *sample* `gen_semantic_model.py` uses for TMDL dtype inference — checking, the actual curated Parquet files never contain that row, so orphan fact rows (KYC decisions referencing a `user_id` absent from Core, ~1.5%

of them) resolve to nothing in the real model. Fixed it properly here: `etl/03_build_curated.py` now appends a genuine `user_key = -1` row to the curated `dim_user.parquet` itself.

Phase 7 — DAX measure library (2026-09-10)

`etl/pbi_measures.py` — 46 measures across 9 folders, `etl/gen_dax_doc.py` → `docs/06_dax_measures.md`.

- **fact_activation's grain was wrong for platform-health measures.** It had been built (Phase 3) as one row per **experiment subject** — 13,577 of 135,000 signups — which made perfect sense for the read-out but left no way to answer "is activation trending up" for the ~90% of signups outside the two experiment windows, exactly what the Executive Overview page needs. Widened `fact_activation` to **one row per signup, platform-wide** (135,000 rows), with a new `is_experiment_subject` column carrying the old meaning forward. Backported the correction into `docs/02_data_architecture.md`, `docs/05_data_model.md`, and the data-dictionary description; re-ran `dq_checks` (rescoped the assignment-integrity check to subjects only, added a grain assertion) — still 0 hard failures.

Phase 8 — Report build, 4 pages (2026-09-10)

Built directly with the `pbir` CLI rather than a bespoke Python driver. The CLI now exposes `add visual --from-json`, `pages background/resize/move`, `add filter`, and `visuals sort/bind` natively — everything the sibling projects' `_drive_pbir.py` + `gen_report_visuals.py` used to hand-roll. `powerbi/build_report.sh` + `_visuals_0N<page>.json` (one manifest per page) replace that pair.

- **No Workspace/Model.SemanticModel to connect to.** `pbir new report -c "Workspace/Model.SemanticModel"` expects a Fabric-published model; this project has no Fabric workspace and never will (it's a local, offline portfolio artifact). The working pattern: `pbir new report ... --thick --no-title` (a throwaway embedded connection), then `pbir report rebind ... --local "../LumaBankActivation.SemanticModel"` to point it at the real sibling `.SemanticModel` folder — a `byPath` connection, exactly what a normal Power BI Desktop PBIP project uses.
- **A pbir/OneDrive interaction bug, hit repeatedly.** `pbir rm` (page or visual) and `pbir pages rename` delete a folder by renaming it to a `.{page,visual}-delete-<id>` staging name and then removing that — and the final removal intermittently failed with `[WinError 5] Acceso negado` on this OneDrive-synced repo, **not transiently**: retrying the identical command 5–6 times over 20+ seconds failed identically every time. A plain PowerShell `Remove-Item -Recurse -Force` on the exact path named in the error succeeded where Python's own delete did not. One rename attempt left `pages.json`'s `pageOrder / activePageName` pointing at a half-deleted page; fixed by hand (a narrow, justified exception to "never hand-edit report JSON" — removing an orphaned id after a botched CLI delete, not authoring new report semantics) and re-validated. `build_report.sh` wraps every delete-sensitive `pbir` call in a retry-with-backoff helper and documents the manual-cleanup fallback in its header. This wasn't independently re-run end-to-end after being written, to avoid re-risking a report that now validates clean.
- **multiRowCard doesn't take a list of fields in one --from-json entry.** The Experiment Readout page's guardrail card needed three unrelated measures as its `Values`; the JSON manifest's `"Values": [...]` array was rejected. Created the visual with one measure, then appended the other two with `pbir visuals bind -a`.
- **Page-level filters carry the story.** The Experiment Integrity Monitor page is filtered to `fact_srm_daily[experiment_phase] = "original"` and the Experiment Readout page to `fact_activation[in_analysis_flag] = TRUE()` — every visual on each page is automatically about the right population without a single measure needing an explicit `CALCULATE` filter for it.

`pbir validate --all`: 4 pages, 26 visuals, 69 fields resolved, 0 warnings, 1 info. Screenshot verification skipped — no live Power BI Desktop session available in this environment; structural validation is the confidence check here.

Phase 10 — Data analysis, the primary deliverable (2026-09-11)

`data_analysis/` — five investigation blocks, dual-track (pandas + DuckDB) throughout, `parity/run_all.py` confirming all of it.

- **01 setup & pre-registration** — the platform's own pre-experiment Day-7 Activation (45.9%) and guardrail baselines; the original 6-week design's MDE (2.41 pp at ~6,750/arm) checked against what the platform history actually said.
- **02 SRM monitoring** — re-derived `fact_srm_daily` independently from the raw assignment log (not trusted from the curated table) and cross-checked 100% agreement. ALERT fires 2026-07-18, two days after the deploy; the cumulative test never crosses 0.001 in the original run (diluted by ten clean days) — confirms why the runbook runs both a cumulative and a trailing-7-day test, not just one.
- **03 root-cause** — the release calendar pins the trigger to 2026-07-16 exactly; 191 contaminated users (4.4%); the fallback allocates 65.6% to treatment overall, skewed from 76.5% treatment in the Southeast down to 35.3% in the North.
- **04 decision analysis** — truncating gives an MDE of 5.4–6.6 pp (worse in real time); excluding only the flagged users gets to 4.8 pp with an unresolved residual bias; neither would have reliably detected the +4.58 pp effect block 05 recovers. Restart was the only option with zero bias.

- **05 re-run readout — +4.58 pp** (95% CI [+2.55, +6.62], $p \approx 1.0e-5$) against a recomputed MDE of 2.92 pp. KYC-rejection guardrail passes non-inferiority cleanly (treatment is directionally *better*); the **flagged-fraud guardrail does not** — the point estimate is small (+0.36 pp) but the one-sided 95% upper bound (+0.89 pp) sits above the +0.5 pp margin. Heterogeneity favours paid_social/influencer as designed, but the formal arm×channel interaction test is not significant ($p = 0.52$) at this sample.

Fix made while building block 03 — a version-string collision

`dim_app_release.is_cache_reset_release` was flagged by matching `version == "4.61.0"` — the hardcoded label for the deploy/hotfix releases. The generator's organic version counter (shared across both platforms and 21 months) coincidentally reached the same number once already, on an unrelated Android release in **2025-10-21**, so the release cross-reference in block 03 initially printed three "cache-reset" releases instead of two, one of them a year early. Fixed by matching on the exact **deploy date** (`released_at == C.DEPLOY_DATE`) instead of the version string — unambiguous regardless of any future numbering coincidence. Re-ran `03_build_curated.py` + `dq_checks` (still 0 hard failures) and regenerated the data dictionary and semantic model.

Deliverables: `deliverables/build_deliverables.py` → `board_summary.{html,pdf}` (decision-first) and `deep_dive.{html,pdf}` (the five findings in full, plus a day-by-day incident timeline from the 2026-07-06 launch through the 2026-08-24→09-20 clean re-run). Recommendation delivered: a **monitored rollout with a fraud circuit-breaker**, not an unconditional launch — the guardrail split (one clean pass, one not established) is a more interesting and more honest result than either "ship it" or "don't ship it" would have been, and it fell out of the data rather than being written in beforehand.

Phase 11 — Portfolio gallery images (2026-09-11)

`portfolio/gen_portfolio_images.py`, adapted from AeroVanti's generator (the closest sibling — same analysis-primary structure, same "frame a real chart + one-line takeaway" pattern) rather than SwiftBite's PBI-screenshot-heavy one, since LumaBank's headline evidence lives in `data_analysis/outputs/figures/`, not the report canvas. Five images, 1600×820, built entirely from real numbers already computed in Phase 10 — no new analysis:

- `01_cover.png` — branded title card (petrol/coral), the four headline numbers.
- `02_the_break.png` — frames `02_srm_break.png` (the daily SRM chart): trailing-7d ALERT on 2026-07-18, cumulative test never gets there.
- `03_fallback_bias.png` — frames `03_fallback_bias.png`: the region-skewed fallback, Southeast 76.5% vs North 35.3% treatment share.
- `04_heterogeneity.png` — frames `05_heterogeneity_forest.png`: the clean re-run's +4.58 pp effect and per-channel breakdown.
- `05_deliverables.png` — the two PDF deliverables, mocked as stacked sheets.

All three source charts (generated in Phase 10) already matched the brand palette without edits — the framing script only adds the header band, kicker, and takeaway strip. Visually verified all five renders before committing.

Phase 12 — Docs bundle & finalize (2026-09-11)

`docs/build_docs_bundle.py`, adapted from AeroVanti's markdown → HTML → headless-Chrome print-to-PDF pattern, re-themed to the petrol/coral palette. This project's doc set stops at `07_visual_identity.md` (no separate `08_data_quality` / `09_report_structure` docs were produced, unlike SwiftBite/AeroVanti — DQ lives in `data/quality/dq_report.md` and report structure was never split into its own doc). Two PDFs:

- `LumaBank-Activation-Documentation.pdf` (2.4 MB) — 00 scope through 07 visual identity, plus the full build walkthrough. 52 pages.
- `LumaBank-Activation-Summary.pdf` (0.9 MB) — scope, business context, data model only.

Rendered the cover page to confirm styling before committing. Added the `## Result` section to the project README (the real numbers, not written until the project was actually done) and closed out the phase checklist.